



Innovation indicators throughout the innovation process: An extensive literature analysis

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ABSTRACT

How to evaluate innovations, especially in the beginning of new product development, is a question constantly posed by academics, managers, and policymakers. One reason for this is that improved front-end decisions greatly affect company performance. To find the answers to this question, this review article analyzes scientific publications on innovation indicators published between 1980 and 2015. The objective of this article is to increase the understanding of the indicator landscape and to complement the various stages of the innovation process with relevant indicators. In doing so, this study categorizes the identified indicators into company-specific and contextual dimensions. Furthermore, this study analyzes the indicators in terms of their potential for ex-ante and ex-post evaluation and investigates the characteristics of relevant publications. The analysis finds that more process rather than product indicators exist in the literature. Current publications emphasize qualitative and indirect indicators but neglect indicators for the early stages of the innovation process. The review identifies 82 unique indicators to evaluate innovations including 26 indicators for the early stages. The results can help managers, researchers, and policymakers to better understand the innovation process and the indicator landscape. However, more concrete indicators are needed to improve front-end innovation decisions.

1. Introduction and motivation

What concrete indicators can be used to evaluate ideas and concepts for innovations before their market entry and after their commercialization, especially during the early stages of the innovation process? This question is repeatedly asked by policymakers, managers, and academic researchers (e.g., Becheikh et al., 2006; Dewangan and Godse, 2014). The increasing number of publications examining innovation indicators and success factors reflects the demand for answers to this question (Becheikh et al., 2006; Freeman and Soete, 2009; Evanschitzky et al., 2012). However, despite the existing research, the indicator landscape still needs to be better understood. Specifically, the front-end of the innovation process requires further clarification (Eling and Herstatt, 2017).

For companies, indicators are indispensable to manage and control the plethora of innovative ideas and concepts that are submitted to them. The defined selection criteria are equally important for an efficient resource allocation and performance evaluation in each phase of the innovation process (Evanschitzky et al., 2012; Dewangan and Godse, 2014). For policymaking practices, it is significant to have accurate indicators to evaluate clearly the proposals of different applicants for innovation projects and to assess the progress of subsidized projects. Improving the evaluation process

of innovations can also help investors to fund new ventures.

Given the significant need to improve the understanding of innovation indicators with a focus on the front-end of the innovation process (OECD, 2005), the interest of this study lies in the indicators and factors behind the innovation performance throughout the innovation process (cf. Birchall et al., 2011; p. 18–19; cf. Klenner et al., 2013, p. 915).

Becheikh et al. (2006) published a systematic literature review on technological innovations in the manufacturing sector from 1993 and 2003. Based on this review, the present study examines the characteristics of innovation indicators, innovation dimensions, and factors. It also complements the various stages of the innovation process with relevant product innovation indicators and process innovation indicators. This complementation leads to a comprehensive overview of all existing ex-ante indicators, and it becomes a starting point for further research. This overview is based on an extensive literature review of scientific publications on the indicators for technological and non-technological innovations published between 1980 and 2015. Therefore, this study covers an extended timeframe and a broader spectrum of industries.

Regarding prior research, the existing (e.g., Montoya-Weiss and Calantone, 1994; Evanschitzky et al., 2012; Storey et al., 2016) and the relationship between innovation and performance of small and

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medium-sized enterprises national culture (e.g., Rosenbusch et al., 2011). Another meta-analysis synthesizes the results on the relationship between the rate of new product development and their antecedents, which are categorized into strategy, project, process, and team (Chen et al., 2010). By contrast, the present study considered a generally broader scope for its research as well as an extended time horizon and a broader type of study. In particular, this study included not only quantitative studies that are synthesized by meta-analyses but also qualitative studies to show the entire indicator landscape.

This analysis can help to advance knowledge on innovation selection indicators by synthesizing the existing results. These results can better channel future studies focusing on the prioritization of innovation projects.

The rest of this paper is organized as follows. First, the related theoretical background and the method used to identify the relevant literature are discussed. Second, the distinguishing characteristics of the relevant publications are presented. Third, the characteristics of innovation indicators throughout the innovation process with a focus on ex-ante and ex-post indicators are analyzed. Finally, the paper consolidates the findings and ends with the main conclusion, implications, and recommendations for researchers, managers, and policymakers.

2. Background literature on innovation indicators and business relevance

The understanding and definitions of *innovation* presented in the existing scientific literature vary greatly from one another, and therefore their use in this study warrants clarification. In this study, *innovation* is defined as “invention plus exploitation,” which is based on Roberts (1998, p. 13) and later used by Dewangan and Godse (2014, p. 536), among others. This definition includes the implementation of a new or significantly improved product, process, or service (OECD, 2005) and the commercialization of innovation (Dewangan and Godse, 2014). Therefore, the term *innovation* applies to a successfully commercialized new idea. For simplicity, this study defines innovation as a term referring to both innovative ideas that are intended to be commercialized in the market and ideas that have already been successfully commercialized.

An *indicator* is considered a measured value that provides information about a specific phenomenon or a status quo. Information can be given in an aggregated form, which facilitates a focused evaluation (Born, 1997). Borrás and Edquist (2013) considered innovation indicators as the source of information from which one can detect problems in the innovation system. This study differentiates among the terms *indicator*, *factor*, and *dimension*. The dimension is understood as the broad field to which the indicator relates (cf. Becheikh et al., 2006). Factor is the more specific field into which the indicator can be categorized. For example, a success factor of the market dimension is customer satisfaction, and an indicator is the number of customer complaints (Fraunhofer-Institut, 2007).

Regarding the stages of the innovation process, ex-ante refers to the front-end of the innovation process. The front-end signifies the generation, screening, and evaluation of ideas and concepts for innovations (Khurana and Rosenthal, 1998; Reid and de Brentani, 2004). Specifically, the front-end is the phase of the first idea until the ideas enter the formal development process, that is, the “go” decision to start the developing process and to commit resources (Eling et al., 2016; Van Oorschot et al., 2018). By contrast, ex-post refers to innovations that have already been introduced into the market, that is, after the market launch.

As noted in the literature on indicators, a number of reviews, aside from the mentioned meta-analyses, present innovation indicators and factors. Nevertheless, the published reviews are insufficient to understand the characteristics of the entire innovation indicator landscape. Thus far, precise selection indicators have not been investigated in adequate detail (Cooper, 1999; Astebro, 2003; Bloch and Bugge, 2013). In particular, ex-ante indicators that can be used in the early stages of the innovation process have been neglected. Instead, researchers have emphasized the influencing factors of innovations (e.g., Balachandra

and Friar, 1997; Fleuren et al., 2014) or indicators that only partially indicate innovations, such as patents (Kleinknecht et al., 2002).

In addition to literature reviews in the indicator research field, prior research also focused on innovation indicators from specific perspectives (e.g., Cooper and Kleinschmidt, 1995; Kerstens-van Drongelen and Cooke, 1997; Verhaeghe and Kfir, 2002; Adams et al., 2006; Chiesa and Frattini, 2009; Cruz-Cázares et al., 2013). The concerned literature focuses on the indirect and direct indicators (Becheikh et al., 2006). Examples of indicators that indirectly and partially evaluate innovations are patents (Hagedoorn and Cloudt, 2003) and research and development (R&D) budget (Flor and Oltra, 2004). Other indicators, such as the number of new product ideas (Cooper and Kleinschmidt, 1993) and the percentage of ideas with a commercialization potential (Dewangan and Godse, 2014), directly evaluate innovations.

From a broader perspective, different understandings of indicators can be found in the literature. For example, Patel and Pavitt (1995) as well as Grupp and Schubert (2010) suggested using composite indicators to measure innovation, as there is no “catch-all” indicator. Other researchers focused on science, technology, and innovation indicators (Freeman and Soete, 2009), and others emphasized input, throughput, and output indicators (e.g., Klomp and Leeuwen, 2001).

Regarding policymaking, a well-known innovation survey using input- and output-oriented indicators is the Community Innovation Survey (CIS) of the European Union (Eurostat) executed by national institutions based on the Commission Implementing Regulation (EU) No. 995/2012 of October 26, 2012 (OECD, 2005; Eurostat, 2015). This questionnaire-based method discusses the technical features and the economic significance of a company's innovative product (e.g., Kleinknecht and Bain, 1993; Cricelli et al., 2016). However, many institutions face the problem of lacking innovation data. Companies are assumed to be unwilling to answer sensitive questions about their innovation processes (Hansen, 1985; Chesnais, 1992). The most well-known manual of international innovation indicators was established by the OECD's “Oslo Manual 2005,” which contains guidelines for gathering and using information about industry innovation activities. A prominent example of innovation measurement is the European Innovation Scoreboard (EIS). The indicators are based on the CIS to compare the innovation performance of EU countries and those of Turkey, Iceland, Norway, Switzerland, the United States, and Japan. The EIS focuses on national and regional comparisons (Hoelscher and Schubert, 2015).

In innovation evaluation in practice, the importance of measuring innovations is increasingly gaining the attention of managers and consultancies. Examples of consulting surveys on innovation measures are the one conducted by The Boston Consulting Group (Andrew et al., 2008, 2010), the McKinsey innovation metric survey (Chan et al., 2008), and the performance management survey by the Business Application Research Center (Bange et al., 2009). Existing surveys demonstrate that rethinking a business's innovation measurement system is crucial (Dewangan and Godse, 2014); this finding is emphasized by practitioners as well. According to the Boston Consulting Group's survey, 74% of managers believed that innovation tracking should be included in central business processes, but only 43% of companies actually measured innovations. Furthermore, 59% of the companies noted that their innovation performance measurement system was not effective (Dewangan and Godse, 2014).

Academic research does not indicate a common overall innovation measurement framework. Moreover, whether the metrics from academic findings are applicable to organizations remains unclear. For example, Adams et al. (2006) claimed that the innovation measurement methods recommended in the research literature seem to be too theoretical. These theoretical indicators are not straightforwardly applicable to businesses (e.g., Adams et al., 2006; Cruz-Cázares et al., 2013). Even a common understanding of the innovation process is missing, as it is quite complex and includes diverse influencing factors (Dodgson and Hinze, 2000; Becheikh et al., 2006). In addition, a measurement strategy to evaluate innovations is lacking (Edison et al., 2013). Consequently, companies face the problem of measuring too few or insignificant data, or they refrain from conducting any innovation

measurement at all (Andrew et al., 2008). Furthermore, organizations disagree on what should be measured.

In sum, measuring newly evolving ideas is a considerable challenge. At the very least, how and what to measure remain unclear when pre-development projects could change in unexpected and diverse ways (Kirchhoff et al., 2013). Another reason for the difficulty in evaluating new ideas may be the unavailable innovation data and methods (Andrew et al., 2008; Edison et al., 2013). The use of indicators is a potential solution for this evaluation problem because it unifies innovation decisions.

Despite innovation indicators having been analyzed in the scientific literature, additional indicators are needed to evaluate the commercial potential of innovations throughout the innovation process. Specifically, ex-ante indicators that can be applied in the early stages of the innovation process are required. To summarize, scholars of applied and theoretical science as well as business practitioners emphasize the importance of innovation measurement in academia and businesses along with the need for a better understanding of the innovation process and the indicator landscape (e.g., Birchall et al., 2011; Edison et al., 2013).

This study has an explorative character. Its main research objective is to increase the understanding of the innovation indicator landscape and to complement the various stages of the innovation process with relevant indicators. Building on existing research, this study focuses on the following classifications of dimensions and indicators that are expected to be significant for a better understanding of the relevant innovation indicators.

1. The identified indicators are categorized into company-specific and contextual dimensions (Becheikh et al., 2006). The specific dimensions are innovation culture, strategy, organizational structure, R&D input and activities, competence and knowledge, financial performance and environment, market, and network.
2. In the innovation indicator literature, the indicators related to the innovative products are published as e.g. the “percentage of ideas found viable for commercialization” (e.g., Dewangan and Godse, 2014).
3. Companies use different criteria at different stages of the innovation process. For example, “time to market” is an indicator for evaluating the length of time it takes from developing a product until the final product launch (e.g., De Felice and Petrillo, 2013).
4. The early stages of the innovation process require different indicators in comparison with the later stages (Hart et al., 2003). In the literature and in practice, the understanding of the front-end of the innovation process is more superficial than that of the later stages (Cooper, 2008; Barczak et al., 2009). However, front-end indicators are significant for the organizational and strategic decision-making process. They also support resource and activity deployment (Hauser and Zettelmeyer, 1997; Hart et al., 2003). Therefore, the current study categorizes product and process indicators into ex-ante and ex-post criteria.
5. Indicators have different characteristics. “Hard” indicators are quantitative in nature, whereas “soft” indicators are qualitative in nature (cf. Freudenberg, 2003, p. 9). Following this differentiation, the present study categorizes the identified indicators into hard and soft criteria.
6. Relevant scientific studies presented indicators that indirectly and directly measure innovations (Becheikh et al., 2006). The current study expands their grouping and identifies the indicators that directly or indirectly influence the success of an innovation. Success is considered as the successful commercialization of innovations in terms of high sales rates (Astebro and Michaela, 2005). A direct indicator is the “percentage of ideas found viable for commercialization” (Dewangan and Godse, 2014) or “future duration of product” (Astebro and Michaela, 2005). Examples of indirect indicators are “newness to business” (Duhamel and Santi, 2012), and “planning and monitoring of the innovation process” (Huergo, 2006). The evaluation used in this study is based on the knowledge of experts working in the innovation management field. Therefore, this study

merely provides an initial categorization that requires further investigation.

7. To display the entire innovation indicator landscape, this study presents the interplay of soft and hard indicators with direct and indirect indicators.

3. Methodology

The study analysis consists of four steps. These steps are explained in further detail in the following subsections.

3.1. Description of the analysis process

The first step is a keyword search in three main databases to gather relevant literature sources on innovation indicators. The databases used are Science Direct, Web of Science, and Scopus. As part of this step, papers from experts working in the innovation management field are also considered. In the second step, the indicators are synthesized to a higher level of dimensions based on the model of Becheikh et al. (2006), as shown in Fig. 1, to categorize the relevant indicators. The dimensions are adapted to the results that have been identified in the referenced literature. Indicators and factors are aligned with the internal and contextual dimensions (Becheikh et al., 2006).

Additionally, each indicator is categorized into “hard” or “soft,” reflecting the quantitative or qualitative aspects, respectively (cf. Freudenberg, 2003, p. 9). In the third part of the procedure, each indicator and each factor are evaluated for whether they directly or indirectly influence the success of an innovation. In this part, success is defined as the enhanced new product performance or the higher success rate of products. An example of a direct indicator is the number of sales of new products, which primarily measures innovation success (e.g., Griffin and Page, 1993). An indirect indicator, such as patents, secondarily influences innovation success (e.g., Acs and Audretsch, 1989).

In the fourth step, the indicators are assigned to a phase of the innovation process (cf. Cooper, 2008). This assigning procedure is important to reflect the entire innovation process and to control for the ex-ante applicability of each indicator. The following stages of the innovation process are based on Cooper (1990) and Hart et al. (2003): *strategy, product definition, product concept, validation phase, production, and market launch*. Steps three and four of this review are conducted by an independent two-stage evaluation process by experts working in the innovation management field. In some cases, no consensus is found on the classification of indicators. The results are then discussed to arrive at an agreement.

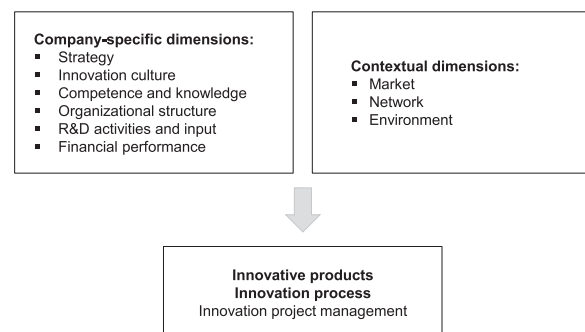


Fig. 1. Adapted framework of Becheikh et al. (2006) to categorize the indicators.

3.2. Article selection process and identification of innovation indicators

The literature review comprises 226 articles, which were selected by a structured keyword search in the above-mentioned databases covering the timeframe of 1980–2015. The following keywords were used

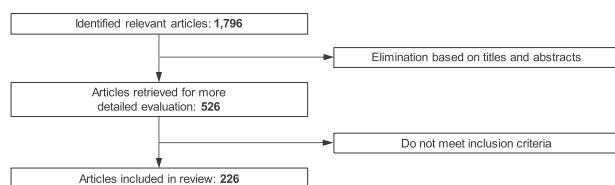


Fig. 2. Flow diagram of the literature review (1980–2015).

to conduct the literature search: innovation indicators, innovation AND indicator, innovation measure or measurement, innovation performance measurement, key or success factors AND innovation, process indicator AND innovation, product indicator AND innovation, determinants of innovation, and factors of innovation.

The period of 35 years was chosen because it catches a wide range of articles and helps to present the evolution of the innovation indicator literature. To proceed systematically, seven inclusion criteria were specified (Alderson et al., 2004).

The material collection comprises only peer-to-peer reviewed articles in scholarly journals to obtain a comparable body of research. To include scientific literature from several research fields and current developments, the search includes a wider area of journals and is not limited to a certain cluster of journals. The inclusion criteria for the innovation indicator literature are as follows:

1. Available in at least one of the three mentioned databases or cited in one of the relevant articles;
2. Comprises one of the keywords of “indicator,” “factor,” or “determinant” of innovation in the title, abstract or full text;
3. Journal publications;
4. Peer-reviewed articles;
5. Published between 1980 and 2015;
6. Articles in English;
7. Articles considering product or process innovations.

The studies were selected as described in Fig. 2. By using the research criteria, 1796 potential articles were identified. Some full texts of relevant articles were not accessible through the mentioned databases. In this case, the authors who published in the journals included in the Social Science Citation Index (SSCI), were asked for their articles directly. Based on the title and abstract criteria, 1270 articles were excluded. The full texts of the remaining 526 potentially relevant articles were screened in more detail. Among the articles, 300 did not meet the inclusion criteria and were thus excluded. In total, 226 articles matched all the inclusion criteria. Either one (or more) of the innovation dimensions or indicators and the factors were presented in the identified study to generate a transparent process. Thus, the review process is replicable and scientific. Consequently, more reliable results were generated from which conclusions could be drawn (Cook et al., 1997). As a basis for further analysis, a Microsoft Excel database was generated and included the following indicator-related information: identified indicators, factors and dimensions, innovation process phase, direct and indirect influence on innovation success, quantitative and qualitative nature as well as publication year, journal, country of author's institution, country of investigation, type of industry, statistical method used for data analysis, qualitative methods to study innovation, and the h5-index of the relevant journal to test for quality of journals.

4. Results

This section presents and discusses the findings from the literature review on innovation indicators. The analysis results based on the relevant publications are first shown, followed by the results of the indicator analysis.

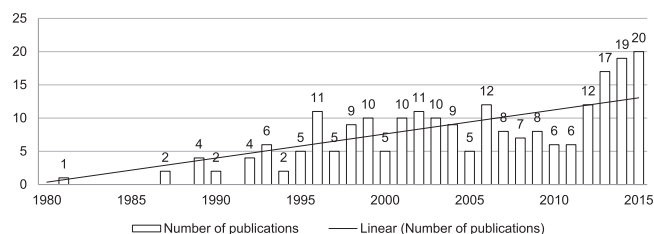


Fig. 3. Publication numbers per year (1980–2015).

4.1. Findings from the publication analysis

4.1.1. Development of innovation indicator publications

Fig. 3 shows the development of publications per year. As shown above, the number of publications between 1980 and 1995 is very low. For 7 of the 16 years, no publication is indicated by the databases. Only in 1993 are there more than six articles published addressing the innovation indicator topic. Beginning at zero or with a low number of articles, the number of publications increases slightly in 1996. Notably, from 2012 onwards, the research focus on innovation indicators increases significantly, with a peak in 2015 of 20 publications. In general, the trend describes an increase in publications over the years from 1980 to 2015. The increase may be due to the series of national innovation surveys that contribute to the European CIS in 1993, 1997, 2001, 2005, 2007, 2009, 2011, 2013, and 2015 (e.g., OECD, 2005; Eurostat, 2015; Behrens et al., 2017).

Table 1 shows the number of publications per journals. Most articles have been published in *Research Policy* (RP), followed by *Technovation*, *Procedia-Social and Behavioral Sciences* (PSBS), and *Journal of Product Innovation Management*. This finding is in accordance with the high impact factors of these leading journals (Journal Citation Reports 2017, Clarivate Analytics, 2018). That is, RP ranks 11th among the world's top journals in Management and first in the Planning & Development category according to the Social Sciences Citation Index (Elsevier, 2018a, referring to Social Sciences Citation Index® by ©Thomson Reuters Journal Citation Reports, 2008).

The impact factor of RP is 3.470 (Elsevier, 2018a). Its higher impact factor compared with other journals implies a high scientific quality of the relevant articles included in the literature review. The impact factor of *Technovation* is 2.243 (Clarivate Analytics, 2018; Elsevier, 2018b). These top journals are followed by other leading journals, such as the *R&D Management*, *Scientometrics*, and *Expert Systems with Applications or Economics Letters*. The high rank of journals is an indication of the high quality of the renowned publications used in this study.

In the h5-index, RP ranks 8th, and *Technovation* ranks 27th. PSBS is not listed under the top journal rankings based on the h5-index (h5-index ≥ 40). The h5-index is used for articles published within the last five years. The index means that h publications have been cited at least five times (each) within the last five years. The list of journals used in the analysis with the available h5-index can be found in the Appendix (Table 10).

4.1.2. Innovation indicator publications at the country level

Table 2 shows the number of publications per country that is investigated in the corresponding study (second column) and per country of the author's affiliated university (third column). If a continent was investigated or if a worldwide investigation has been conducted, the study was excluded because identifying one specific country would be impossible.

The United States, Germany, the United Kingdom, and Spain are the leading countries in both the investigated country and the country of the author's affiliation. However, the UK, Spain, and Germany interchange in the second, third and fourth places, depending on the view of the country (country of investigation or country of the author's affiliated university). The majority of these authors investigated their country of affiliation and infrequently studied an additional country

Table 1

Number of publications per journal (1980–2015).

Journal	Number of publications per journals
Research Policy	36
Technovation	17
Procedia - Social and Behavioral Sciences	11
Journal of Product Innovation Management	9
R&D Management	8
Scientometrics	6
Creativity and Innovation Management	3
Economics Letters	3
Expert Systems with Applications	3
Journal of Business Venturing	3
Research Evaluation	3
Research Technology Management	3
Review of Industrial Organization	3
Technology Analysis & Strategic Management	3
Academy of Management Journal	2
Applied Economics	2
IEEE Transaction on Engineering Management	2
Industrial Marketing Management	2
Industry and Innovation	2
International Journal of Innovation Management	2
International Journal of Technology Management	2
Journal of Cleaner Production	2
Journal of Communication	2
Journal of Engineering and Technology	2
Journal of Marketing Research	2
Management Science	2
Small Business Economics	2
Strategic Management Journal	2
Structural Change and Economic Dynamics	2
Technological Forecasting & Social Change	2
The Economic Journal	2
World Patent Information	2

Note: This table shows only publications that occurred more than two times between 1980 and 2015.

(e.g., Baptista and Swann, 1998; Mattes et al., 2006; Alegre and Chiva, 2008; Lecerf, 2012; Pekovic et al., 2015). Another reason for the high number of publications in the United States and the United Kingdom may be that several high-ranking universities are located in these countries, such as the London Business School and University of Oxford in the United Kingdom and Harvard University, Stanford University, and Massachusetts Institute of Technology in the United States (e.g., Top universities, 2015). Although China is only in the 8th or 10th place, the publications investigating data on Chinese industries have been increasing since 2009. Canada, the Netherlands, and Turkey are countries that are usually investigated in comparison with India, Portugal, or South Korea, for example.

In terms of the countries and the investigated industry, research interest in the United States focuses heavily on patents (Acs and Audretsch, 1989; Dror, 1989; Lanjouw and Schankerman, 2004; Mattes et al., 2006; Belenzona and Pataconi, 2013). In the United Kingdom, interest lies in the qualification and experience of employees (Hoffman et al., 1998; Romijn and Albaladejo, 2002) and in the interactions with external partners, such as universities, suppliers, or customers (Romijn and Albaladejo, 2002), among others. Research interest in Germany is varied, as authors focus on external influences on innovation (Brem and Voigt, 2009), project delay (Feurer et al., 1996), patent stocks (Czarnitzki and Kraft, 2004), and other areas. Regarding industry, German authors analyze the branches of manufacturing (Czarnitzki and Kraft, 2004) and technology-based services (Brem and Voigt, 2009). Spanish authors show particular interest in the tourism and agriculture sectors, such as food and beverage (March-Chordà et al., 2002; Nieves et al., 2014), as well as in ceramic tile industry (Flor and Oltra, 2004) and start-up branch. These interests are an indication of innovation and

Table 2

Number of publications per country where the survey was conducted and per country of the author's affiliated university (1980–2015).

Country	Number of publications per country where the survey was conducted	Number of publications per country of the author's affiliated university
USA	26	37
Spain	15	19
Germany	14	22
United Kingdom	11	22
Canada	10	13
Turkey	10	12
Netherlands	8	17
France	8	4
Italy	7	9
China	5	9
Finland	4	5
Japan	4	3
Switzerland	4	5
Australia	3	6
Belgium	3	8
Denmark	3	0
Greece	3	0
Sweden	3	5
Brazil	2	4
Columbia	2	2
Croatia	2	0
Ireland	2	0
Malaysia	2	2
Norway	2	4
Poland	2	0
Portugal	2	2
Slovenia	2	2
South Korea	2	2
Taiwan	2	5
Thailand	2	0
India	0	3
Hungary	0	2
Israel	0	2
Singapore	0	2

creative activity (Sáez-Martínez et al., 2014). Compared with other sectors, the manufacturing sector is generally presented more often (Huerger, 2006; Vega-Jurado et al., 2008; Gonzalez-Benito et al., 2015). For example, Turkey has shown a growing interest in the innovation indicator research since 2001. In this country, industries such as yacht building and software development (Koc, 2007) as well as small companies (e.g., Bayarçelik et al., 2014) have been investigated.

4.1.3. Innovation indicator publications at the industry level

The following chapter analyzes the industries that have been investigated in the used literature. The analysis is based on the NACE (Nomenclature statistique des activités économiques dans la Communauté européenne) classification (Eurostat, 2008). The manufacturing industry, which has been studied the most, has a score of 74%. The huge amount of research on innovation indicators in the manufacturing industry underlines the interest in the evaluation criteria for innovations at the company level. The remaining industries account for 1–4% of the relevant publications. These other studied industries are usually situated in the service sector. The data show a shift from the manufacturing to the service industry. This shift could have resulted from specific trends, such as digitization, big data movement, and the need for companies to focus on services for customers to be successful in the market (Hipp and Grupp, 2005; Chandler, 2015). However, the number remains quite low when compared with that of the manufacturing industry. Publications related to “diverse industries” and articles that did not focus on a specific branch are excluded in the analysis because the corresponding industries are not well described in these articles.

4.1.4. Methods used to study innovation

Fig. 4 indicates that, neglecting other and unspecified methods, regression analysis is applied the most frequently at 27% (53 times) in comparison with the other methods of relevant publications.

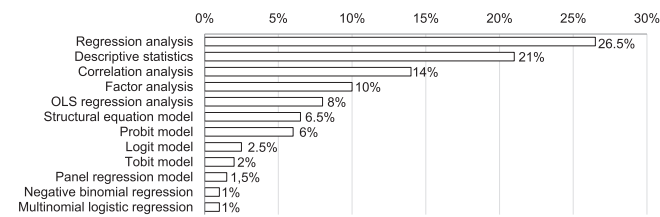


Fig. 4. Statistical and econometric methods used to study innovation (1980–2015).

“Regression analysis” includes all types of regression analyses, which are not listed here individually. The other data analysis techniques to investigate innovation are descriptive analysis (21%) and correlation analysis (14%). The presented methods show that factor analysis (10%), ordinary least squares regression (8%), structural equation model (6.5%), and probit model (6%) are the frequently used methods. To sum up, the relevant articles discuss different methods to study innovations. Most of these identified methods are used individually or jointly. However, to present the methodological landscape, each method is pointed out separately in this study. Articles that did not use a common quantitative method are not listed. Conceptual models are rarely used (only two times) (Cooper and Kleinschmidt, 1993; Brown and Eisenhardt, 1995).

4.2. Synthesis of the literature: company-specific and contextual indicators and factors

To classify the broad range of indicators, a framework is set up based on the model of Becheikh et al. (2006), as previously mentioned. The framework was adapted and refined for this article, as shown in Fig. 1, and it is complemented by the numbers shown in Fig. 5.

After screening the relevant literature, the indicators are categorized into company-specific and contextual dimensions. These dimensions determine the innovation process and the resultant innovation product. The identified indicators are classified according to these dimensions. As previously mentioned, an example of an innovation factor of the market dimension is customer satisfaction, and an indicator is the number of customer complaints (Fraunhofer-Institut, 2007).

Multiple factors related to internal and external elements affect the ability of companies to implement innovations successfully (Rothwell et al., 1974). Therefore, two categories are set up. First, company-specific dimensions include those that are particular to a company, such as culture or structure, and that affect the organizational innovation behavior (e.g., Souitaris, 2002a). Second, contextual dimensions are related to a company and its surrounding environment (Becheikh et al., 2006). The latter dimension is based on the contingency approach (Lawrence and Lorsch, 1967; Woodward, 1970), which defines a company as an adaptive system that reacts to the surrounding environment in terms of its strategy, structure, and culture (Becheikh et al., 2006). In total, 11 dimensions are determined (without innovation project management). The dimensions, their attached factors and indicators, and their influence on innovations and the innovation process are explained in detail in the next section. Nonetheless, the focus lies on the product and process indicators, which are also presented in the subsequent section. The study takes a broad view on the indicators, followed by a more focused view on the product and process indicators.

Only a few publications examine the precise indicators. Many authors analyze dimensions or factors that influence innovations instead of using actual indicators. Out of the 800 relevant dimensions and indicators found in the literature review, only 371 indicators are mentioned in the relevant publications. This number represents the number of scientific publications that investigate indicators. Regarding this calculation, the dimensions and indicators are counted twice (or more). As for the unique indicators, 82 product and process indicators and

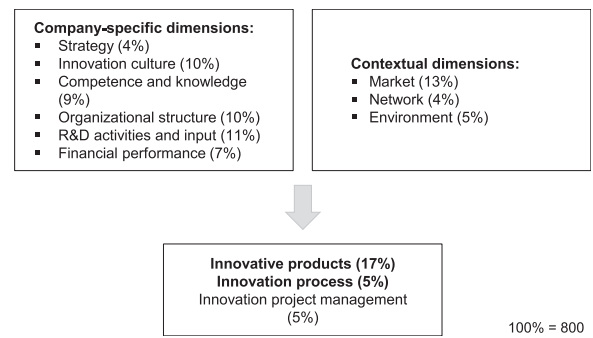


Fig. 5. Dimension framework for synthesizing the innovation indicators based on the dimension framework of Becheikh et al. (2006).

factors to evaluate innovations throughout the innovation process are identified in the relevant literature.

4.2.1. Company-specific dimensions: Culture, strategy, structure, and R&D activities

Table 3 shows examples of the factors and indicators that are categorized under the company-specific dimensions identified in the innovation literature between 1980 and 2015. The full list of categories and indicators of the company-specific dimensions is found in the Appendix (Table 11).

The following subsection explains the company-specific dimensions in detail.

4.2.1.1. Strategy and vision. A company's strategy defines the future activity fields to achieve long-term company goals, and it is the baseline for defining a company's organizational innovation goals (Porter, 1987; cf. Souitaris, 2002a; Astebro and Michaela, 2005). At only 4%, the strategy dimension is presented the least (together with the network dimension). A category of this dimension is innovation strategy (Adams et al., 2006; Kamasak, 2015). The number of newly created innovative opportunities is a strategy indicator (Hittmar et al., 2015).

4.2.1.2. Innovation culture. Integrating innovation into the company culture is an important means to achieve success and to foster innovation capabilities (Bullinger et al., 2007). The beliefs and values of a company influence the risk tolerance, personal development, and innovation activities of employees and their motivation to develop and implement new ideas (Menzel et al., 2007). The innovation culture (Fig. 5) is mentioned comparatively often. It constitutes 20% along with organizational structure. The indicators to measure the innovation culture are the percentage of leaders trained in creativity techniques (Chiesa et al., 1996) and the amount of time managers spent with the management of an innovation compared with their usual tasks (Hittmar et al., 2015), among others.

4.2.1.3. Competence and knowledge. The competences and knowledge of a company's employees are crucial resources for new ideas and innovation projects. The individual competence is the ability to implement knowledge into actions to achieve the defined goals. At 9%, this dimension (Fig. 5) is presented quite often. Innovation-oriented learning (De Medeiros et al., 2014) is a category in this field.

4.2.1.4. Organizational structure. The organizational structure regulates how rules, hierarchies, and responsibilities are established, controlled, and coordinated. Regarding the previously identified publications, the relation between business size and innovation is investigated comparatively often in the reviewed literature. However, different results have been published. On one hand, small companies seem to have an advantage in the management of their innovations (Rothwell,

Table 3

Examples of categories and indicators categorized under the company-specific dimensions (1980–2015).

	Factor	Indicator	Relevant references
Innovation culture	Innovation culture of organization	–	Bayarçelik et al., 2014; Slater et al., 2014; Naranjo-Valencia et al., 2015
	Creativity	Percentage of leaders trained in creativity techniques, atmosphere	Chiesa et al., 1996; Ayob et al., 2012; Edison et al., 2013; Yang et al., 2015
	Company's entrepreneurial orientation/spirit	–	Al-Mubarak et al., 2015; Gonzalez-Benito et al., 2015
	Top management support	Amount of time managers spent with innovations compared to normal tasks	March-Chordà et al., 2002; Graner and Mißler-Behr, 2013; Hittmar et al., 2015
	Openness of company towards change and innovation	Number of external ideas/generated with customers	Enkel et al., 2005; Ogawa and Piller, 2006; Lenfle, 2008; Dewangan and Godse, 2014
Strategy	Resistance to change	–	Veugelers and Cassiman, 1999
	Innovation strategy	–	Adams et al., 2006; Kamasak, 2015
	New product strategy	–	Huang et al., 2004
	Strategic fit of innovation	Number of newly created innovative opportunities	Griffin and Page, 1993; Hittmar et al., 2015
	Willingness to take risks	–	Aiman-Smith et al., 2005; Astebro and Michaela, 2005; Wan et al., 2005; Salomo et al., 2007; Escalfoni et al., 2011; Murro, 2013
Knowledge and competence	Innovation-oriented learning	Number of managers trained in the methods and tools of innovation	Kerssens-van Drongelen and Cooke, 1997; Banerjee, 1998; Astebro and Michaela, 2005; De Medeiros et al., 2014; Hittmar et al., 2015
	Openness towards knowledge	–	Caloghirou et al., 2004
	Internal knowledge resources, experiences and background of founder/managers	Use of internal and external knowledge and information sources	Caloghirou et al., 2004; Sawang, 2011; Kamasak, 2015; Kato et al., 2015
Organizational structure	Business data, organizational factors	Size of the company, Geographic location of the company, Age of company, External and internal growth, Formal structure	Wan et al., 2005; Huergo, 2006; Krasniqi and Kutllovci, 2008; Koouba et al., 2010; Tohidi and Jabbari, 2012; Wang, 2012; Frey et al., 2013; Slater et al., 2014; De Fuentes et al., 2015; Kamasak, 2015; Pekovic et al., 2015
	Flexibility, rapid adaptation to customers	–	Wu et al., 2002; Krasniqi and Kutllovci, 2008; Suwannaporn and Speece, 2010
	Internal communication	–	Lester, 1998; Suwannaporn and Speece, 2010
	Good team structure together with appropriate leadership	Accountable, dedicated, supported cross-functional teams with strong leaders	Cooper and Kleinschmidt, 1993; Griffin and Page, 1993; Cooper, 1999; Hollemann et al., 2009; Weiss et al., 2011
	Team satisfaction	–	–
Research and development activities and input	Willingness to invest in innovation/R&D, willingness to conduct new research projects = sufficient amount of investment, financial resources dedicated to innovation	R&D expenditure/investment	Avermaete et al., 2004; Caloghirou et al., 2004; Katz, 2006; Chiesa et al., 2009; Belitz et al., 2011; Weiss et al., 2011; Tohidi and Jabbari, 2012; De Felice and Petrillo, 2013; Edison et al., 2013; Makkonen and van der Have, 2013; Dewangan and Godse, 2014; De Medeiros et al., 2014; Kim, 2014; Cavdar and Aydin, 2015; De Fuentes et al., 2015
	Research activities	Average expenditure per selected idea	–
	–	Percentage of sales related to new projects	–
	–	Share of research budget from total company budget	–
	–	Innovation expenditure	–
Financial innovation performance	Share of technology transfer	–	–
	Return on investment in innovation	–	–
	R&D costs/revenue in %	–	–
	Profit margin measures	–	–
	New-to-market and new-to-business sales	–	–
	Percentage of innovations that met financial benefit projections	–	–

1986; Bughin and Jacques, 1994). On the other hand, large companies are more likely to invest in innovative projects because they can allocate greater R&D resources than small firms (Becheikh et al., 2006). Furthermore, studies found that size determines the relationship between the management of innovations and the marketing of innovations (Gonzalez-Benito et al., 2015). In addition to the lack of capital, small companies partly face the challenge of information deficits, such as missing details about innovation policy instruments, technical information, and highly qualified employees (Kleinknecht, 1989). One reason for the diverging results in the analyzed literature may be that these publications investigated the relationship between business size and innovation in different contexts (i.e., different countries, periods, and methods). Nevertheless, this dimension has been investigated frequently, with a factor of 10%. An indicator for this dimension is business size (Huergo, 2006), and team satisfaction is considered a success factor (Griffin and Page, 1993).

4.2.1.5. R&D activities and input. R&D activities and input are related to the financial situation of a business (Beneito, 2003) and the availability of resources. Resources in this sense refer to employees, technology, tangible assets (e.g., machinery, tools, and materials), time spent with an innovation, and investments made to develop and realize innovative products. Different results are found in the relevant publications. For example, R&D personnel ratio (internal expertise) has a strong positive effect on product and process innovation, and process innovation is also influenced by R&D intensity (R&D investment) (Song and Oh, 2015). R&D intensity seems to be influenced by knowledge and technology transfer activities (Arvanitis et al., 2008). Arvanitis et al. (2008) defined the innovation performance of companies as the R&D intensity and the number of sales of new products. In the examined literature, R&D budget (Flor and Oltra, 2004), or business investment (Sosnowski, 2014), is listed as one of the main indicators to measure innovative activity and recognize innovating corporations. This indicator might be considered to be used to

measure innovation because of its availability (Jacobsson et al., 1996). However, the data on R&D expenses are given in a limited form, as many companies have strict confidentiality policies to secure their competitive advantage (Kleinknecht, 1993). For small businesses, formal R&D expenditures are difficult to capture as the R&D budget may be designated as other expenses (Kleinknecht, 1987). R&D expenditures are accompanied by newly gained knowledge and by the building of networks among different organizations, research institutions, or universities (Cavdar and Aydin, 2015). Raymond and St-Pierre (2010) found that “process innovation,” such as the improvement of a production method, mediates the effect of R&D on product innovation (Raymond and St-Pierre, 2010). In general, high investments in innovations lead to an improved innovation performance (De Fuentes et al., 2015). Even though the R&D indicators are a good representation of organizational innovativeness (Romijn and Albaladejo, 2002), they only provide insights into the innovation input and not into the specific innovativeness of a company (Godin, 2002). Therefore, R&D measures innovation indirectly. Moreover, not all innovations are based on R&D (Becheikh et al., 2006). Generally, R&D and financial indicators are represented well in the relevant publications (11%), and this situation might be due to the fact that these indicators are based on numbers that are more accessible. One category of the R&D dimension is willingness to invest (Atebro and Michaela, 2005). An indicator for this category is R&D expenditures (Adams et al., 2006).

4.2.1.6. Financial innovation performance. Financial performance is defined as the earnings of a business through the sale of innovative products in the market. Financial performance is found to be the third lowest dimension in the relevant research publications. One reason for this result is that, although innovation plays an important role in the success of a business, the actual success that is based on innovation is difficult to capture. Examples of indicators in this field are return on investment with innovations (Kim, 2014) and new-to-market and new-to-business sales (Caird, Hallett, and Potter, 2013).

4.2.2. Contextual dimensions: Network, market, internationalization, and environment

Table 4 presents the examples of categories and indicators classified according to the contextual dimensions that are identified in the innovation literature between 1980 and 2015. The full list of categories and indicators of the contextual dimensions is found in the Appendix (Table 11).

The following subsections explain the contextual dimensions in detail.

4.2.2.1. Network.

Generally, a company network includes the

collaboration with external partners, suppliers, and institutions that are important for the company's innovation capability. The network dimension (Fig. 5) is presented the least often at 4% (together with the strategy dimension). Examples of network categories and indicators are R&D alliances or co-patents (Adams et al., 2006), cooperation with universities, research centers, competitors (Keizer et al., 2002), and customer and supplier relationships (Kamasak, 2015). From the articles published after 2006, an increase in the openness of companies toward innovation can be identified (e.g., Alcaide-Marzal and Tortajada-Esparza, 2007).

4.2.2.2. Market. Market focus plays an important role when aiming for product success (e.g., Avermaete et al., 2004; Atebro and Michaela, 2005; Bullinger et al., 2007). At the market level, demand and supply determine the success of a business (Freeman, 1979; Zahra, 1993; Atebro and Michaela, 2005). The market dimension is the second most frequently investigated dimension at 13%. Examples of market categories and indicators are purchase intention rate (Griffin and Page, 1993), sales share of new or highly improved services (%) (Hollenstein, 2003), and export activities (Martinez-Ros, 1999).

4.2.2.3. Environment. Multiple factors related to internal and external elements affect the ability of organizations to implement innovations successfully (Rothwell et al., 1974). The environment as the surrounding of a business is rarely investigated (5%). The environment indicators are the number of innovative businesses (Alcaide-Marzal and Tortajada-Esparza, 2007) and new venture companies (Al-Mubarak et al., 2015), among others.

The emphasis of the next section is on product and process indicators and factors to increase the understanding of indicators throughout the innovation process.

4.3. Findings from the process and product indicator analysis

4.3.1. Innovation process

As described previously, the innovation process (Fig. 6) is usually complex (Dodgson and Hinze, 2000). To evaluate the indicators relevant to the mentioned ex-ante view to analyze future success, the innovation process is divided into different stages based on the frameworks of Cooper (1990) and Hart et al. (2003). The first stage includes the innovation strategy as a preparatory step. To categorize the indicators into the different stages and to reduce the complexity, this study assumes the innovation process to be linear. Nevertheless, note that the linear portrayal is dynamic and iterative, with diverse feedback loops to adapt to the

Table 4

Examples of categories and indicators classified according to the contextual dimensions (1980–2015).

	Factors	Indicator	Relevant references
Market	Market demand	Demand growth in the industry Duration of demand	Freeman, 1979; Zahra, 1993; Crépon et al., 1998; Atebro and Michaela, 2005
	Maintenance and expansion of market share, growth	Market share, position and share	Cooper, 1981; Griffin and Page, 1993; Palmberg, 2006; Mendes Luz et al., 2015
	Competitor analysis/monitoring of competitors	New product introduction vs. competition	Lukas and Ferrell, 2000; Ivanova and Avasilcăi, 2014
	Customer satisfaction	Customer complaints Response time to customer requests Delivery reliability and/or speed Customer retention rate	Atebro and Michaela, 2005; Enkel et al., 2005; Chiesa et al., 2009; Sawang, 2011; De Felice and Petrillo, 2013; Dewangan and Godse, 2014; Fleuren et al., 2014
Network	Internal and external collaboration	R&D alliances	Flor and Oltra, 2004; Blindenbach-Driessen and van den Ende, 2006; Belitz et al., 2011; Oyeleran-Oyeyinka and Adebowale, 2012; Caird et al., 2013; De Medeiros et al., 2014
		Knowledge and technology transfer activities with research institutions and/or institutions of higher education	
Environment	Innovative environment	Number of innovative businesses/new venture start-ups	Alcaide-Marzal and Tortajada-Esparza, 2007; Al-Mubarak et al., 2015
	Stakeholders	–	Bloch and Bugge, 2013
	Political driving forces (e.g. government stability, taxation policy) and support by specific policies and programs	–	Brem and Voigt, 2009; Bloch and Bugge, 2013; Pervan et al., 2015



Fig. 6. Simplified innovation process based on Cooper (1990) and Hart et al. (2003).

market requirements (e.g., Kline and Rosenberg, 1986).

- 1) **Strategy:** In this phase, the product's strategy is defined to achieve a unique selling proposition.
- 2) **Product definition:** With this next step, the product itself is defined and the market requirements are identified to meet customer expectations (Lester, 1998). Ideas for innovative products are generated and evaluated.
- 3) **Product concept:** The product concept is created on the basis of the product definition to coordinate and start the validation and production phases. The potential costs and required resources are considered according to the business case calculations of the innovation idea. The development of the product begins.
- 4) **Validation phase:** Prototypes are developed and tested to validate and fulfill the diverse requirements.
- 5) **Production phase:** When the innovation is produced on a small scale and the processes are approved, the production of a (pre-) series starts, and new products are subsequently produced in high volumes.
- 6) **Market launch and commercialization:** The innovations are ready to be produced in series. The final products are introduced into the market with a communication and marketing strategy to achieve the highest sales figures. For the market launch, different indicators (e.g., the number of products launched) can be found in the selected literature to measure the innovations (Hittmar et al., 2015).

The innovation process itself is considered a success factor; that is, the quality of the process affects new product development (Cooper and Kleinschmidt, 1995). This result is in accordance with the model of Utterback and Abernathy (1975), which assumes a mutual relation between the product and the process innovation rate. The model explains the technological developments, and it differentiates among three phases. From the process view, the stages are uncoordinated, segmental, and systematic. From the product perspective, the stages are performance maximization, sales maximization, and cost maximization. In the first phase, a high product innovation rate accompanies a low process innovation rate. The process is flexible, and a high number of product variations exist. The innovation competition is high, and the market share remains low. The second phase is characterized by a decreasing product innovation rate and an increasing process rate. At this point, the process flexibility decreases, and higher sales quantities are produced. Price pressure comes with a growing competition, resulting in efficiency enhancements and first standardizations. The third phase refers to the further decreasing product innovation rate that reaches stagnation and a decreased innovation process rate. This phase signifies a higher product quality, and companies compete for price leadership. High process standardization implies high product standardization. Product variations at this level of standardization are cost intensive. In sum, product and process innovations influence each other (Utterback and Abernathy, 1975).

4.3.2. Innovation process indicators

In comparing the process and product indicators shown in Fig. 5, both process indicators and factors are investigated less frequently (5%) than product indicators and factors (17%). However, in comparing the number of unique indicators, their amounts are almost the same. The relevant unique indicators and factors to evaluate the innovation process and products are presented in Tables 5, 6. The tables also show the characteristics of the indicators, that is, whether the indicators are of soft, hard, indirect, or direct nature.

Additionally, the innovation process needs to be managed, and its management mainly involves the planning, supervision, and controlling of the innovation process. The management of the innovation process is essential because it affects the success of the innovation process (Cooper, 1999). Therefore, this category is also included in Table 5.

Examples of process indicators are the time it takes to develop the next generation of the product (Ivanova and Avasilcăi, 2014), innovative activities (Therrien and Mohnen, 2003), and the gap between plans and action (Kim, 2014).

4.3.3. Product indicators

According to the research literature, several factors are essential for achieving a high product performance. For example, customers are crucial for the success of new products. Specifically, customer expectations should be met by adding new or problem-solving functions, thereby satisfying customer needs (Chiesa et al., 2009; Duhamel and Santi, 2012; Dewangan and Godse, 2014). This argument implies that the advantage of an innovative product should be visible to a customer and the handling of the product's functionality should be as intuitive as possible (Cooper, 1999; Astebro and Michaela, 2005).

Standardization is also important to achieve a good technological performance (e.g., Blind, 2001; Chiesa et al., 2009). A common indicator for innovation measurements is the number of patents or citations based on patent data (22 times), which is in accordance with the results of other studies (e.g., Adams et al., 2006). Patents show a high potential as a measurements tool. However, using patent data as an innovation indicator has some limitations (Kleinknecht et al., 2002). Therefore, the following aspects must be considered: 1) patents protect inventions and not innovations, 2) not all innovations are patented, and 3) different propensities of patenting behavior are dependent on a company's strategy and sectors (Arundel and Kabla, 1998). In addition to the third point, the economic value of a patent (Griliches, 1979; Pakes and Griliches, 1980) or the motives to patent (Blind et al., 2006) should be considered when using these data for measuring innovation. Patents are used for strategic purposes, such as to block other patents with unused patents or to receive licensing fees (Torrise et al., 2016). Furthermore, from a strategic viewpoint, patents are used to improve a company's position compared with its competitors or in negotiations with licensees or partners. Patents also play a role as an indicator for performance measures and incentives for R&D personnel (Blind et al., 2006). Furthermore, the motives to patent and the number of sleeping (unused) patents are positively correlated with company size (Blind et al., 2006; Torrise et al., 2016). Despite these limitations of patents as an innovation indicator, patents represent new technologies. As patents are correlated with innovation activity (Acs et al., 2002), they can indicate innovations. They also provide helpful hints to understand the innovation processes within international research interactions (Stek and Geenhuizen van, 2015).

The important product innovation indicators are the number of new product ideas (Cooper and Kleinschmidt, 1993), future duration of products (Astebro and Michaela, 2005), and number of project definitions with business approval (Tipping et al., 1995). To evaluate ideas for innovations, these indicators may play an important role in the product definition and concept phase. Furthermore, a precise, stable, and early product definition before development starts (Cooper, 1999) is important for the subsequent development of a product, and thus it could be used as an indicator. Moreover, the innovation portfolio balance may be essential from a strategic point to achieve high success rates with innovations (Adams et al., 2006; Kim, 2014). After market launch, the new product performance or the success rate of new products becomes an important indicator. It can be measured by the percentage of innovations that met the financial profit, the profitability of newly listed products, and the number of products launched within the last three years (Griffin and Page, 1993; Chiesa et al., 2009; Tohidi and Jabbari, 2012; Edison et al., 2013; Ivanova and Avasilcăi, 2014; Kim, 2014; Hittmar et al., 2015), among others. Product advantage (Cooper,

Table 5

Process innovation indicators and factors identified in the relevant literature and categorized under the innovation process (1980–2015).

	Indicators and factors	Relevant references	s/h	i/d
Product definition	Time to develop next generation	Ivanova and Avasilcăi (2014)	h	d
	Diversity of idea-generation process	Kim (2014)	s	i
	Time for idea generation	Lester (1998)	h	i
	Idea-tanks management	Yang et al. (2015)	s	d
	Idea generation and its management	Koc (2007), Escalfoni et al. (2011)	s	d
	Motivational factors in the work environment to generate ideas	Foss et al. (2013)	s	d
	Product development planning and process	March-Chordà et al. (2002)	s	i
	Business planning	Aiman-Smith et al. (2005)	s	i
	Involvement of product champions/% of projects for which an effective project champion can be identified on the project team	Tipping et al. (1995), Blindenbach-Driessen and van den Ende (2006)	h	i
	Number of on-going innovations	Hittmar et al. (2015)	h	d
Product concept	Innovation activity	Therrien and Mohnen (2003)	s	d
	Explicit project selection	Blindenbach-Driessen and van den Ende (2006)	s	i
	Design orientation, such as, number of designers on the company's staff, or source of design	Alcaide-Marzal and Tortajada-Esparza (2007)	h	i
	Making business cases	Blindenbach-Driessen and van den Ende (2006)	s	i
Validation phase	Innovative projects (in-house)	Caird et al. (2013)	s	i
	Detailed project tactical plan	Lester (1998)	s	i
	Formal ratification by management	Fleuren et al. (2014)	s	i
	Implementation of innovation activity	Belitz et al. (2011)	s	d
	Motivational factors in the work environment to implement ideas	Foss et al. (2013)	s	d
	Gap between plans and action	Kim (2014)	h	d
Production phase	Cost of production/new product developments	Astebro and Michaela (2005)	h	i
	Ease of manufacture	Griffin and Page (1993)	s	i
	Manufacturing efficiency and productivity	Griffin and Page (1993), Pachico (1996), Palmberg (2006), Han et al. (2009), Damijan et al. (2012), Hittmar et al. (2015)	h	d
	Increase in production capacity and flexibility	Mendes Luz et al. (2015)	s	d
Market launch	Internal process lead time	Han et al. (2009)	h	i
	Number of optimized production processes that the company used for its products	Tohidi and Jabbari (2012)	h	i
	Use of new technology for production of new products	Tohidi and Jabbari (2012)	s	i
	Planning and manufacturing system	Yang et al. (2015)	s	i
	Reassessment efforts: update and redirect project plans and keep team members aligned	Lester (1998)	s	i
	Time to implement the innovation	Fleuren et al. (2014)	h	d
	Process time	Sawang (2011)	h	i
	Time from identification of a customer product need until beginning of commercial	Tipping et al. (1995)	h	d
	Time to market	Adams et al. (2006), Chiesa et al. (2009), De Felice and Petrillo (2013), Edison et al. (2013), Hittmar et al. (2015)	h	d
	Building lead time	Han et al. (2009), Sawang (2011)	h	i
	Labor productivity	Sawang (2011)	h	i
	Number of improved processes	Hittmar et al. (2015)	h	i
	Percentage decrease in the cost of innovative processes and products	Hittmar et al. (2015)	h	d
	Percentage of project milestones achieved	Tipping et al. (1995)	h	i
	Duration of introduction of product	Escalfoni et al. (2011)	h	d
	Indented purpose achievement of innovation	Astebro and Michaela (2005)	s	i
Innovation process management	A well-planned, adequately resourced and proficiently executed launch	Cooper (1999)	s	d
	Determinants related to facilities that are needed to implement the innovation	Fleuren et al. (2014)	s	i
	Total cost of all commercially successful projects divided by the number of commercially successfully projects	Tipping et al. (1995)	h	d
	Project efficiency in relation to cost and time	Chiesa et al. (2009)	h	d
	Measurement of time	Griffin and Page (1993)	h	i
	Project delay	Feurer et al. (1996)	h	d
	Rate of received approval on time	Han et al. (2009)	h	i
	Quality of execution of the activities that comprise the innovation process	Cooper and Kleinschmidt (1993)	s	i
	High-quality new product process	Cooper and Kleinschmidt (1995)	s	i
	Efficient processes like tough go/kill decision points or gates	Cooper (1999)	s	i
	New product development process/process management itself	Lester (1998), Slater et al. (2014), Raja and Wei (2015)	s	i
	Heavyweight project management	Blindenbach-Driessen and van den Ende (2006), Lenfle (2008)	s	i
	Percentage of projects in the total portfolio going through a defined project management system with defined milestones	Tipping et al. (1995)	h	i
	Clear goals and milestone measurements	Lester (1998)	s	i
	Percentage of completion of objectives at the expected milestone date	Tipping et al. (1995)	h	i
	Planning and monitoring of the innovation process	Huergo (2006)	s	i
	Way in which the new product development process is formalized	Graner and Mißler-Behr (2013)	s	i
	Common understanding of the process for new product development	Lester (1998)	s	i
	Procedural clarity	Fleuren et al. (2014)	s	i
	Project ownership/empowerment (= support and freedom)	Tipping et al. (1995)	s	d
	Flexibility and agility, such as centralization of decision-making	Koberg et al. (1996)	s	i
	Feedback effects in-between innovation decisions	Martinez-Ros (1999)	s	i
	Responsibility	Gana (1992)	s	i
	Project leader	Flipse et al. (2013)	s	i
	Replacement for leaving staff	Fleuren et al. (2014)	s	i
	Monitoring of the innovation process and the hiring of personnel with special skills for technological tasks	Huergo (2006)	s	i
	Number of meetings	De Felice and Petrillo (2013)	h	i

s = soft (qualitative indicator)|h=hard (quantitative indicator).

d=direct influence on innovation success|i = indirect influence on innovation success.

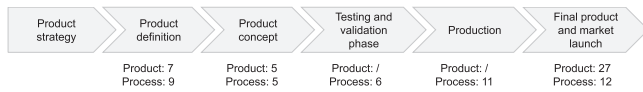
Table 6

Product innovation indicators and factors identified in the relevant literature and categorized under the innovation process (1980–2015).

	Indicators and factors	Relevant references	s/h	i/d
Product definition	Number of new product ideas or suggestions	Cooper and Kleinschmidt (1993), Chiesa et al. (1996), Chiesa et al. (2009), Dewangan and Godse (2014)	h	d
	Percentage of ideas found viable for commercialization	Dewangan and Godse (2014)	h	d
	Precise, stable and early product definition, before development begins	Cooper (1999)	s	i
	Synergy potential/dependency on other products	Griffin and Page (1993), Astebro and Michaela (2005)	s	i
	Innovation portfolio balance	Adams et al. (2006), Kim (2014)	s	i
	Newness to company and novelty of product	Cooper (1981), Duhamel and Santi (2012)	s	d
Product concept	Future duration of product	Astebro and Michaela (2005)	s	d
	Intellectual property rights, in particular patents, citations, applications, licenses	Basberg (1987), Acs and Audretsch (1989), Dror (1989), Griliches (1990), Trajtenberg (1990), Brouwer and Kleinknecht (1991), Grupp (1992), Jaffe et al. (1992, 1993), Shane (1993), Littell (1994), Geisler (1995), Tipping et al. (1995), Macdonald and Lefang (1998), Veugelers and Cassiman (1999), Katila (2000), Blind (2001), Acs et al. (2002), Beneito (2003), Hagedoorn and Cloodt (2003), Hollenstein (2003), Therrien and Mohnen (2003), Czarnitzki and Kraft (2004), Flor and Oltra (2004), Lanjouw and Schankerman (2004), Astebro and Michaela (2005), Hipp and Grupp (2005), Adams et al. (2006), Lin and Lu (2006), Mattes et al. (2006), Alcaide-Marzal and Tortajada-Esparza (2007), Tseng and Wu (2007), Gittelman (2008), Chiesa et al. (2009), Ejermo (2009), Yunwei et al. (2009), Buesa et al. (2010), Guan and Chen (2010), Bayarçelik and Taşel (2012), Adams et al., 2013; Belenzona and Pataconi (2013); De Rassenfosse et al. (2013), Dereli and Altun (2013), Karvonen and Kässi (2013), Makkonen and van der Have (2013), Dewangan and Godse (2014), Kim (2014), Sosnowski (2014), Thoma (2014), Akis (2015), Al-Mubarak et al. (2015), Cavdar and Aydin (2015), Dang and Motohashi (2015), Hittmar et al. (2015), Raja and Wei (2015), Rocha et al. (2015), Roper and Hewitt-Dundas (2015)	h	i
	Power law distributions of patents	O'Neale and Hendy (2012)	s	i
	Technological significance	Astebro and Michaela (2005)	s	i
	Technical standards and standardization degree	Blind (2001), Chiesa et al. (2009)	h	i
	Number of project definitions with business/marketing approval	Tipping et al. (1995)	h	d
	–	–		
	–	–		
	–	–		
	–	–		
Validation phase	–	–		
Production phase	–	–		
Market launch	New product performance/ success rate of new products, e.g. percentage of innovations that met financial profit estimates, profitability of newly listed products, number of products launched (last three years)/output quantity	Griffin and Page (1993), Han et al. (2009), Dewangan and Godse (2014), Hittmar et al. (2015)	h	d
	Number of new or improved products within a certain period, or number of new products divided by total number of products offered	Griffin and Page (1993), Therrien and Mohnen (2003), Hipp and Grupp (2005), Alcaide-Marzal and Tortajada-Esparza (2007), Chiesa et al. (2009), Tohidi and Jabbari (2012), Edison et al. (2013), Ivanova and Avasilcăi (2014), Kim (2014), Hittmar et al. (2015)	h	d
	Counts of new product announcements, introduction of technological innovations to the market	Beneito (2003), Mendes Luz et al. (2015)	h	d
	Successful idea implementation, such as number of ideas successfully turned into products or shared ideas submitted to successful ideas	Edison et al. (2013), Hittmar et al. (2015)	h	d
	Identification of innovations by business managers	Flor and Oltra (2004)	s	i
	Assessments by experts	Kleinknecht and Reijnen (1993)	s	i
	Perceived value	Griffin and Page (1993)	s	d
	Survival rate	Griffin and Page (1993), Al-Mubarak et al. (2015)	h	d
	Straightforwardness (how easily can the customer learn the correct use of the innovation?)/customer familiarity with the innovation and specialization	Cooper (1981), Astebro and Michaela (2005), Duhamel and Santi (2012)	s	d
	Product advantage: differentiated, unique benefits, superior value for the customer, recognizably of advantage	Cooper (1981, 1999), Astebro and Michaela (2005)	s	d
	Degree of innovativeness	De Brentani (2001)	s	d
	Appearance	Astebro and Michaela (2005)	s	d
	Awareness of content of innovation	Fleuren et al. (2014)	s	d
	Product quality and reliability, such as customer evaluation/defect rate assessment	Griffin and Page (1993), Tipping et al. (1995), Baldwin and Johnson (1996), Sawang (2011), De Felice and Petrillo (2013), Edison et al. (2013), Mendes Luz et al. (2015)	h	d
	Comparative functionality	Astebro and Michaela (2005)	s	i
	Compatibility, correctness, completeness	Astebro and Michaela (2005), Fleuren et al. (2014)	s	i
	Complexity of innovation	Fleuren et al. (2014), Findik and Beyhan (2015)	s	i
	Societal benefit	Astebro and Michaela (2005), Chiesa et al. (2009)	s	i
	Opportunity window	Cooper and Kleinschmidt (1987)	s	i
	Launch quality and quantity	Cooper and Kleinschmidt (1993)	h	d
	Duration of product life cycles	Hittmar et al. (2015)	h	d
	Sustainability/eco-efficiency of product	Kobayashi (2006)	s	d
	Extending the range of offered products	Mendes Luz et al. (2015)	h	i
	Numerical rankings of a business' product by a given customer divided by that customer's ranking of the best competitive product	Tipping et al. (1995)	h	i
	% of sales protected by patents, trade secrets, other exclusive know-how	Tipping et al. (1995)	h	i
	Licensed technology use	Güngör and Gözlü (2012)	h	i
	Trademarks	Schmoch and Gauch (2009), Flikkema et al. (2014)	s	d

s = soft (qualitative indicator) | h = hard (quantitative indicator).

d = direct influence on innovation success | i = indirect influence on innovation success.



Product indicators: 39; process indicators: 43, in total: 82

Fig. 7. Indicators and factors throughout the innovation process (1980–2015).

1999) is important to attain competitive advantage, which shows unique benefits and a superior value to customers. Table 6 presents the complete list of identified product innovation indicators.

4.3.4. Ex-ante versus ex-post indicators

In this section, the identified indicators and factors are classified into the stages of the innovation process. Fig. 7 shows the number of identified unique indicators throughout the innovation process.

As shown in Fig. 7, the product definition phase presents 16 indicators, and the product concept phase (ex-ante) shows 10 indicators and factors. The validation and production phases even show a complete lack of product innovation indicators. The market launch phase (ex-post) presents the highest number of indicators and factors (39). In total, 82 indicators are found in the relevant publications.

The early stages of the innovation process show fewer indicators and factors than the end of the innovation process (market launch). One reason for this may be that vague ideas and concepts are difficult to evaluate in the beginning of the process (Kim and Wilemon, 2002). This may also be the reason why more qualitative indicators are represented in the early phases (Fig. 8), as capturing quantities at the beginning of the innovation process is more difficult. Intellectual property rights, mostly patent data, are frequently investigated in the research literature as a way to measure innovation. Depending on a company's expertise, patents are applied in the concept phase to ensure the freedom to operate (Ernst, Conley, and Omland, 2016). Moreover, indicators for “strategy” are lacking in the identified literature.

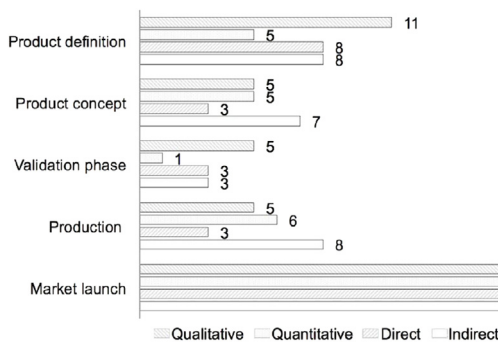


Fig. 8. Indicators throughout the innovation process categorized into qualitative, quantitative, direct, and indirect indicators (1980–2015).

The results highlight that the number of indicators for the later stages exceeds the number of indicators for the early stages of the innovation process. The results also point out that the more progressed the innovation process is, the higher the number of product indicators. This outcome can be explained by the fact that with the finalized product, more data are available to evaluate the product.

The ex-ante view plays a significant role in the evaluation of decision-making processes, which can be linked to ex-post results (e.g., Poti and Cerulli, 2011). Therefore, focusing on the ex-ante and ex-post indicators is important. Ex-ante decision making refers to phases one to three of the innovation process, which precede the validation phase and in which the product is defined and conceptualized. Changes are still easier to implement in the early phases than in the validation stage or subsequent stages. Innovative ideas need to be evaluated in terms of their potential success in the future. Although potential success is difficult to assess, decisions about which ideas or projects should be

Table 7
Most frequently used ex-ante and ex-post indicators and factors (1980–2015).

Product indicators	Ex-ante indicators	Selected references	Ex-post indicators	Selected references
Patents/patent applications		Chiesa et al. (2009), Makkonen and van der Have (2013), Rocha et al. (2015), Roper and Hewitt-Dundas (2015)	Straightforwardness (how easily can the customer learn the correct use of the innovation?)	Astebro and Michaela (2005)
Customer focus		Raja and Wei (2015)	Count of new product announcements	Beneito (2003)
Number of ideas		Chiesa et al. (2009), Hittmar et al. (2015)	Number of new products	Chiesa et al. (2009)
Percentage of ideas found viable for commercialization		Dewangan and Godse (2014)	Product advantage, differentiated, unique benefits, superior value for the customer	Cooper (1999)
Future duration of product, technology significance		Astebro and Michaela (2005)	Degree of innovativeness	De Brentani (2001)
Dependency on other products		Astebro and Michaela (2005)	Number of improvements in existing products	Edison et al. (2013)
Novelty to the company		Duhamel and Santi (2012)	Success rate of new products and rate of survival on market	Griffin and Page (1993)
		Lester (1998)	Shared ideas submitted and successful ideas	Hittmar et al. (2015)
		Therrien and Mohnen (2003), Hittmar et al. (2015)	Time taken in turning an idea into a product or market launch	Adams et al. (2006); Edison et al. (2013)
Time for idea generation		Koc (2007)	Rate of suggestions implemented	Chiesa et al. (2009)
Number of on-going innovations/ innovation activity		Escalfoni et al. (2011)	Total cost of all commercially successful projects divided by the number of commercially successful projects	Tipping et al. (1995)
Idea generation		Cooper (1999)	Number of improved processes	Hittmar et al. (2015)
Idea management		Aiman-Smith et al. (2005)	Time from identification of a customer product need until commercial sales	Tipping et al. (1995)
Precise, stable and early product definition before development begins		Ivanova and Avastici (2014)		
Business planning				
Time to develop next generation				

Table 8

Overview of the indicators categorized into quantitative and qualitative as well as direct and indirect (1980–2015).

	Hard	Soft	Direct	Indirect	Total
Combinations	x		x		22
		x	x		20
	x			x	23
		x		x	41
Total	45	61	42	64	

soft = qualitative indicator|hard = quantitative indicator.

direct = direct influence on innovation success|indirect = indirect influence on innovation success.

focused on need to be made during the early process stage. Nowadays, in the validation phase, projects are halted because of the lack of technical or production feasibility. Table 7 shows the most frequently used ex-ante and ex-post indicators and factors.

Although some ex-ante indicators can be identified, more indicators for the ex-ante evaluations of innovations in policymaking and organizational contexts seem to be necessary to better estimate the potential of innovations. This overview shows that ex-post indicators mostly refer to the innovation performance in terms of quantitative values. However, exactly assessing the innovation success ex-post remains difficult. In addition, the benchmark of indicator values deserves further research to benefit from the application of indicators in practice. Nevertheless, each institution or company should predefine the values that it aims to reach per observation unit. In other words, the indicator benchmark depends on the corresponding industry. For example, the same values of indicators for the software branch might not as important as they are for the automotive industry to the same extent.

4.3.5. Characteristics of product and process indicators: qualitative and quantitative as well as direct and indirect

In this subsection, the research results about the product and process indicators are outlined in terms of their quantitative and qualitative nature (Hoelscher and Schubert, 2015), as illustrated in Fig. 8. Moreover, the direct and indirect nature of an indicator, which illustrates its influence on an innovation's success, is shown. As mentioned previously, this study considers the term “success” to mean high sales figures (Astebro and Michaela, 2005).

As shown through a comparison of the different phases of the innovation process, the product definition phase shows more qualitative (soft) than quantitative (hard) indicators. This result could be a sign that using quantitative values to assess vague innovative ideas at the

beginning of the innovation process would be difficult. At this point, indirect and direct indicators show the same numbers. Remarkably, in the product concept phase, the quantitative and qualitative indicators are found to be equally distributed in the selected literature, whereas the indirect indicators exceed the direct indicators. The validation phase shows a higher number of qualitative than quantitative indicators and factors. In the production phase, quantitative and indirect indicators exceed the number of direct indicators. The higher indirect rate of indicators could exist because the indicators that monitor the process itself are more important during the production phase than the indicators related to the success of the product. The market launch phase shows the highest number of indicators. In this phase, the quantitative and direct indicators are the most represented indicators reviewed literature, closely followed by the qualitative and indirect indicators (Fig. 8).

Tables 8, 9 provide further insights into the indicator characteristics. The results reveal that the number of unique process indicators slightly exceeds the number of unique product indicators. This outcome emphasizes the notion that the innovation process itself influences the innovative product (Utterback and Abernathy, 1975). A high-quality innovation process accompanies a high-quality product. However, the differences between the process and the product indicators are only minor. Indicators help to manage the process and to achieve the defined goals. The general indicators for evaluating the innovation management process are listed here separately because categorizing these indicators into only one phase is impossible.

The results show that more qualitative (61) than quantitative (45) indicators are mentioned in the relevant literature. One reason for this finding is that innovations are difficult to depict as concrete values, especially at the beginning of the innovation process. Additionally, there are fewer direct (42) indicators than indirect (64) indicators. Therefore, the indicators that directly influence the innovation success are underexplored to some extent. This result underlines the need for further studies on direct indicators. The qualitative indicators may be sufficient to evaluate innovations in the early stages of the innovation process because the actual number of innovations is rare in this initial phase. Nevertheless, further investigations are required to identify the relevant qualitative indicators that can be used by researchers, managers, and policymakers. To summarize, the qualitative and indirect indicators are investigated more than the quantitative and direct indicators. However, this needs to be further analyzed and tested.

Specifically, process indicators show more quantitative and indirect indicators than product indicators, as shown in Table 9. The reason for this finding may be that more general project management indicators and factors are known for the innovation process. The qualitative

Table 9

Overview of the product and process indicators and factors categorized into hard and soft as well as direct and indirect nature throughout the innovation process (1980–2015).

	Product				Process				Total
	Soft	Hard	Direct	Indirect	Soft	Hard	Direct	Indirect	
Product definition	5	2	4	3	6	3	4	5	16
Product concept	2	3	1	4	3	2	2	3	10
Validation phase	0	0	0	0	5	1	3	3	6
Production	0	0	0	0	5	6	3	8	11
Market launch	15	12	16	11	3	9	5	7	39
General innovation management	17	7	4	20					24

soft = qualitative indicator|hard = quantitative indicator.

direct = direct influence on innovation success|indirect = indirect influence on innovation success.

factors and indicators are equally represented with regard to the product and process indicators. Direct product indicators and factors are mentioned more than the direct process indicators. For product indicators, a large number of indicators and factors have already been published, but more specific product indicators are needed to evaluate innovations.

In comparing the process stages, the results highlight that the number of soft indicators exceeds the number of hard indicators in the early stages of the innovation process. This outcome can be explained by the fact that at the front-end, the exact number of ideas is difficult to generate. Moreover, the number of indirect indicators exceeds the number of direct indicators in the early stages of the innovation process.

Regarding the characteristics of innovation indicators, a change to a higher number of soft indicators is observable. These soft indicators refer to non-technological and technological innovations as well as a broader spectrum of industries (including service industries) and a wider timeframe. However, further investigation on indicators is required to generalize the results.

Although a notable set of indicators is known, the actual indicators to assess the potential of innovation remain mostly missing probably because of the described lack of particular data necessary to evaluate innovations. Especially at the beginning of the innovation process, estimating the future success of innovations, particularly without sufficient data, is challenging. Furthermore, the innovations and their influencing factors may vary in unforeseeable ways. Therefore, evaluating innovations in the early stages of new product development is problematic.

Accessible data are used more often than information that is difficult to obtain. Availability can also be a reason why some indicators are used more frequently than others. Therefore, the further the process of innovation progresses, the more indicators are available. This leads to the following question: Are additional qualitative indicators required for the beginning of the innovation process? This question is in accordance with the OECD results, suggesting that some ideas can be covered by quantitative indicators, whereas others should be evaluated by qualitative ones (OECD, 2005).

Nevertheless, in practice, measuring newly evolving ideas is a considerable challenge at the very least because it is unclear how and what to measure when the necessary conditions tend to change in unexpected and diverse ways (Kirchhoff et al., 2013).

5. Conclusion and implications

This study analyzed the characteristics of innovation indicators throughout the innovation process. By considering existing studies on the indicators, determinants, and categories of innovations, this work aimed to explore the ex-ante and ex-post evaluation criteria of non-technological and technological innovations. The main reason for investigating the indicators was to assure the decision quality of innovations. A clear indicator definition and clustering of innovations can help companies to manage ideas and innovations throughout the innovation process (Kerssens-van Drongelen and Cooke, 1997).

The literature review confirms the variety of indicators and factors of innovations. Furthermore, this study shows that different evaluation indicators, as mentioned in the selected literature, can complement the innovation process. Ex-ante indicators, such as number of ideas, percentage of ideas with commercialization potential, dependency on other products, and customer orientation, are identified for the early stages. Ex-post indicators, such as number of new products, product advantage, success rate of new products, and time span between the

identified customer need and the final product launch, are identified for the product and process evaluation after the market introduction.

The research literature also suggests an uncertainty among researchers about innovation indicators in general. In particular, the analysis shows the lack of appropriate indicators for the early stages of the innovation process. Front-end indicators are important for the operations (i.e., for policymakers and managers) and research departments. A great number of easily obtainable data, such as data on patents, are frequently investigated. However, some aspects, such as the motives to patent, should be regarded when using them as innovations indicators (e.g., Becheikh et al., 2006; Blind et al., 2006).

The analysis results indicate that more process indicators than product indicators are mentioned in the existing academic literature. Current publications emphasize qualitative and indirect indicators for the early stages of the innovation process. The results also point out that the more progressed the innovation process is, the higher the number of product indicators. This outcome could be due to the type of data available on the finalized product; these data are more accessible than data on innovative ideas. Specifically, the review identifies 82 innovation indicators and factors, 26 of which can be used in the early stages of the innovation process. The indicators related to product strategy are underrepresented. By comparing the characteristics of innovation indicators, including non-technological and technological innovations, a broader spectrum of industries, and a wider timeframe, this study observes a movement toward a larger number of soft indicators. This finding can be explained by the increasing focus on service industries. Nevertheless, further investigation on the indicators is required to generalize the results.

Despite the high number of well-known indicators and factors, concrete indicators to evaluate innovations are difficult to identify in the selected literature. Furthermore, transferring the known indicators from theory to practice is problematic. One reason for this problem is the lack of data that can be applied to the indicators. Instead of precise criteria, publications emphasize factors that do not measure the status quo of innovations in an exact manner. Nevertheless, these factors are important because they have a positive or negative impact on the innovation outcome, depending on the identified publications. This raises the question of whether appropriate indicators are actually available to assess the potential of innovative ideas or whether they are influencing factors rather than indicators. Furthermore, the benchmarking (i.e., the value) of the indicators may also be interesting for practitioners to analyze in future studies. For example, in the view of the authors, defining the value of indicators to facilitate their application in practice may be significant. The indicator benchmark may be dependent on the corresponding industry and the innovation process.

6. Limitations and future research

This study presented an extensive overview of the indicators and factors of innovations retrieved from an intensive review of published academic literature dealing with these topics. The results can be used to better channel upcoming research on innovation decisions. Consequently, future research will enable policymakers and companies to improve their decision-making processes with regard to innovation. On one hand, the review expands the innovation indicator landscape by type of industry and methods and extends it over a longer period of 35 years. On the other hand, the diversity of methods used to study innovation makes generalizing and combining existing results even more difficult. A limitation of this study is that the quantitative and qualitative as well as the direct and indirect evaluations of the indicators were conducted on the basis of a two-stage evaluation process by

experts working in the innovation management field only. The results may be subjective rather than objective despite their proficiency of these experts. Another limitation is the high number of indirect indicators that are considered to be secondarily linked to the innovation success combined with the low numbers of indicators identified in the early stages of the innovation process. This leads to the recommendation that investigating precisely the direct and indirect effects the indicators have in the early stages on future innovation success is necessary. Therefore, further analysis of the connection between the criteria and success to derive front-end indicators is suggested. A more in-depth understanding can help practitioners to focus on the most promising innovations and to foster them. Another limitation is that the publication sample is based on a restricted range of selected keywords. Other relevant articles on the topic of innovation evaluation might not have been included as a result of this method. The relevant literature used in this study comprises a wide and extensive spectrum of publications. Nevertheless, several factors and indicators are occasionally vaguely presented in these publications and are thus difficult to apply in practice. In this context, the study also recommends limiting the literature analysis to a narrower frame to enable a greater in-depth analysis of the possibilities for evaluating innovations at the beginning of

the innovation process. Owing to the increasing number of publications on innovation indicators as well as the opportunities for big data to be a source for further innovation indicators, future examinations of this topic are still required.

Accordingly, this study proposes to examine the use of the identified indicators and other options to ex-ante evaluate innovations in practice. Specifically, it suggests analyzing the relevance and the applicability of foresight evaluations of indicators. Note that front-end indicators are difficult to investigate, but they are highly needed at the same time. Therefore, it may be helpful for future studies to analyze the possible use of indicators to evaluate innovative ideas regarding their success potential.

In sum, this research provides the foundation for further research on innovation measurement frameworks that are applicable in the front-end.

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Appendix

See [Tables 10, 11](#) here.

Table 10

List of journals used in the analysis with available h5-index from Google Scholar Metrics, survey date: April 1, 2016.

Journal	h5-index	h5-median
NBER WORKING PAPERS	163	233
PLOS ONE	161	210
Journal of Financial Economics	113	173
Energy Policy	98	125
Expert Systems with Applications	91	107
Health Affairs	82	123
Academy of Management Journal	78	115
Research Policy	77	117
Strategic Management Journal	71	113
Journal of Cleaner Production	67	95
Management Science	67	100
World Development	67	94
The Economic Journal	66	99
Journal of Business Venturing	63	110
Applied Mathematics and Computation	57	76
Industrial Marketing Management	56	79
Journal of Marketing Research	54	85
The Leadership Quarterly	53	75
International Journal of Project Management	52	70
Journal of Economic Literature	52	138
Journal of the Academy of Marketing Science	51	97
Omega International Journal of Management Science	50	65
International Journal of Nursing Studies	49	66
Journal of Systems and Software	49	60
Small Business Economics	48	70
Technovation	48	71
Journal of Product Innovation Management	47	70
Management Decision	47	64
Trends in Food Science & Technology	47	73
Computers & Industrial Engineering	46	66
International Journal of Hospitality Management	46	71
Scientometrics	46	58
International Journal of Management Reviews	42	74
Journal of Communication	40	74

Table 11
Categories and indicators organized into company-specific and contextual dimensions identified in the innovation literature (1980–2015), full list of Tables 3 and 4.

Category		Indicator	Relevant references
Innovation culture	Company-specific dimensions		
	Organization and culture	-	Jung et al., 2003; Adams et al., 2006; Bayarçelik et al., 2014; Slater et al., 2014; Naranjo-Valencia et al., 2015
	Innovation culture	Creative approaches in problem solving	Chiesa et al., 1996; Ayob et al., 2012; Edison et al., 2013; Al-Mubarak et al., 2015; Yang et al., 2015
	Creativity	Creative environment (subjective assessment)	
		Percentage of leaders trained in creativity techniques	
		Atmosphere of innovation among staff	
	Business' entrepreneurial orientation	-	Gebhardt and Pohlmann, 2013; Al-Mubarak et al., 2015; Gonzalez-Benito et al., 2015
	Entrepreneurial spirit	Compatibility of innovation with current attitudes and workflow/ personal routine	Astebro and Michaela, 2005
	Good attitude towards innovation	-	Belitz et al., 2011
	Attitudes toward science and technology	-	Belitz et al., 2011
	Social innovation climate and trust	-	Kerssens-van Drongelen and Cooke, 1997; De Brentanti and Kleinschmidt, 2004
	Learning organization	-	De Brentanti and Kleinschmidt, 2004; Caird et al., 2013
	Support of new ideas	-	
	Organizational support for innovative-projects in-house	-	Feurer et al., 1996; de Brentanti, 2001; Enkel et al., 2005; Dewangan and Godse, 2014
	Innovation capability	Percentage of ideas generated in new domains	
	Open and highly innovative new product culture within the company	Number of externally generated ideas	
	Openness to new fields	-	De Felice and Petrillo, 2013
	Staff satisfaction	-	Edison et al., 2013; Sosnowski, 2014
	Presence of innovation champions	-	Escalfoni et al., 2011
	Way in which ideas are managed	-	Flor and Oltra, 2004
	Acceptance of/participation in publicly funded innovation support programs	-	Flor and Oltra, 2004
	Identification of innovations using information provided by experts in the sector	-	
	Top management support	The amount of time managers/senior management spent with innovation compared to normal operational tasks	Henard and Szymanski, 2001; March-Chordà et al., 2002; Graner and Mißler-Behr, 2013; Kim, 2014; Hittmar et al., 2015
	Senior management support	Number of identified problems (product, process...)	Hittmar et al., 2015
	Innovation analysis	-	Kasa, 2015
	Measures for innovation potential: e.g., push and pull technologies, culture etc.	-	Lenfle, 2008
	Overlapping problem-solving cycles (also called concurrent engineering)	-	Parthasarathy and Hammond, 2002
	Reward system	-	Lenfle, 2008
	Openness of company towards change and innovation	Number of external ideas/number of implemented ideas	
	Customer involvement	The integration of customers and suppliers into product development	
	Perception of cost/risk related to innovation	Number/percentage of ideas generated in collaboration with customers	Enkel et al., 2005; Ogawa and Pillar, 2006; Dewangan and Godse, 2014
	Resistance to change	-	Veugelaers and Cassiman, 1999
	Belief that innovation is important	-	Veugelaers and Cassiman, 1999
	Willingness to exchange ideas	-	Wan et al., 2005
	Tolerance for innovation failures	Number of failures	Wan et al., 2005
			Yang et al., 2015

(continued on next page)

Table 11 (continued)

Strategy	Category	Indicator	Relevant references
Strategy	Types of strategy	-	Adams et al., 2006; Bullinger et al., 2007; Kamasak, 2015; Yang et al., 2015
	Innovation strategy	-	Slater et al., 2014
	Product launch strategy	-	Huang et al., 2004
	New product strategy	-	Koschatzky et al., 2001; Souitaris, 2001
	Marketing strategies	-	Hollemann et al., 2009
	Team directed strategies	-	Michie and Sheehan, 2003
	Human resource management	-	Kerssens-van Drongelen and Cooke, 1997
	Focus	-	Chiesa et al., 2009
	Long-term focus	-	Griffin and Page, 1993
	Cost reduction	-	Hittmar et al., 2015
	Strategic fit of innovation	-	Souitaris, 2002a; Raja and Wei, 2015
	Strategic activities	-	Salomo et al., 2007
	Strategic orientation	-	Flipse et al., 2013
	Strategic planning	-	Pekovic et al., 2015
	Goal stability	-	Aiman-Smith et al., 2005; Astebro and Michaela, 2005; Wan et al., 2005;
	Social and ethical aspects	-	Salomo et al., 2007; Escalfoni et al., 2011; Murro, 2013
	Internal strategic behavior	-	Suwanaporn and Speece, 2010
Knowledge and competence	Riskiness of the business	-	Chiesa et al., 1996; Adams et al., 2006; Idris and Trey, 2011; Nieves et al., 2014
	Strategy and planning	-	Kerssens-van Drongelen and Cooke, 1997; Banerjee, 1998; Astebro and Michaela, 2005; de Medeiros et al., 2014; Bayarçelik et al., 2014
	Knowledge management	-	Banerjee, 1998; Martinez-Ros, 1999; Avermaete et al., 2004; Chiesa et al., 2009; Güngör and Gözli, 2012; Mendes Luz et al., 2015; Kato et al., 2015
	Innovation-oriented learning	-	Hittmar et al., 2015
	Information	-	Jacobsson et al., 1996; Uzun, 2001; Romijn and Albaladejo, 2002; Avermaete et al., 2004; Flor and Oltra, 2004; Astebro and Michaela, 2005; Alcaide-Marzal and Tortajada-Esparza, 2007; Chiesa et al., 2009; Beltz et al., 2011; Idris and Trey, 2011; Sawang, 2011; Güngör and Gözli, 2012; Tohid and Jabbari, 2012; Caird et al., 2013; De Felice and Pettilo, 2013; Romero, 2014; Yague et al., 2014; Hittmar et al., 2015; Yang et al., 2015
	Sources of information	-	Caloghirou et al., 2004
	Use of external sources of information	-	Kleinknecht and Reijnen, 1993; Link, 1995; Coombs et al., 1996; Santarelli and Piergiovanni, 1996; Walker et al., 2002; Flor and Oltra, 2004; Mendonca et al., 2004; Fagerberg et al., 2007; Katz, 2006; Chen et al., 2010; Caird et al., 2013; Dereli and Altun, 2013; Sosnowski, 2014; Cavdar and Aydin, 2015; Hittmar et al., 2015; Rocha et al., 2015
	Innovation training	-	
	Acquiring/accumulation of knowledge, knowledge improvement, competence acquisition	-	
	Support for education of employees	-	
	Number of managers having training in the methods and tools of innovation	-	
	Extent of internet usage for tasks	-	
	Researchers (input)	-	
	Purchase of information and information updates	-	
	Number of new competencies for innovation	-	
	Enterprise training	-	
	Qualification of employees	-	
	Staff training	-	
	Company's investment in know-how	-	
Knowledge and competence	Identifying innovations in technical and specialized journals (LBIOs)	-	
	Publication count/number of publications	-	
	Number of awards, publications	-	
	Scientific impact (citations/paper)	-	
	Increasing publications/articles	-	
	Publications by the company as the champion in innovation	-	
	Open excellent attractive research	-	
	International scientific co-publications	-	
	Top 10% most cited scientific publications worldwide	-	
	Organizational support for public/private research co-publications	-	
Law and regulation knowledge		-	
		-	

De Medeiros et al. 2014

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Table 11 (continued)

Category	Indicator	Relevant references
Organizational structure	Technological capabilities derived from in-house R&D	Daellenbach et al., 1999; Souitaris, 2002b; Caloghirou et al., 2004; Oerlemans and Pretorius, 2006; Vega-Jurado et al., 2008; Sawang, 2011; Kamasak, 2015; Kato et al., 2015; Yang et al., 2015 Liu et al., 2014 Yang et al., 2015 Yang et al., 2015 Yang et al., 2015 Chang, 2003 Baldwin and Johnson, 1996; Hoffman et al., 1998; Martínez-Ros, 1999; Romijn and Albaladejo, 2002; Avermaete et al., 2004; Bayarçelik et al., 2014 Sosnowski, 2014 Bertschek and Entorf, 1996; Koberg et al., 1996; Love et al., 1996; Evangelista et al., 1997; Romijn and Albaladejo, 2002; Koc, 2007; Bishop and Wiseman, 1999; Blind and Grupp, 1999; Davenport and Bibby, 1999; Sørensen and Stuart, 2000; Belderbos, 2001; Sternberg and Arndt, 2001; Fritsch and Meschede, 2001; Chang, 2003; Uzun, 2001; Greve and Seidel, 2003; Jung et al., 2003; Hipp and Grupp, 2005; Wan et al., 2005; Huergo, 2006; Zdunczyk and Blenkinsopp, 2007; Kivimaa, 2008; Krasniqi and Kutllovi, 2008; Koutba et al., 2010; Tohidi and Jabbari, 2012; Wang, 2012; Frey et al., 2013; De Fuentes et al., 2015; Slater et al., 2014; Kamasak, 2015; Pekovic et al., 2015 Fleuren et al., 2014 Wu et al., 2002; Krasniqi and Kutllovi, 2008; Suwannaporn and Speece, 2010; Yang et al., 2015 Al-Mubarak et al., 2015 Astebro and Michaela, 2005 Fleuren et al., 2014 Fleuren et al., 2014 Fleuren et al., 2014 Yang et al., 2015 Escalfoni et al., 2011 Fleuren et al., 2014 De Brentanti and Kleinschmidt, 2004 Hemphälä and Magnusson, 2012 Fleuren et al., 2014
	Internal knowledge resources, employee knowledge in terms of educational background, experiences and background of founder/managers	
	Integrative capabilities	
	Design capability	
	Service capability	
	Ability of information collection and information management	
	Networking ability to cooperate with buyers, suppliers and external organizations	
	Skills and experience of managers	
	Qualification and experience of employees	
	Intellectual assets	
	Business data/organizational factors	
	Size of the business	
	Large company's advantages: use and effectiveness of technological management mechanisms	
	Geographic location of the company	
	Age of company	
	Scale	
	Field	
	Company's characteristics (size) and external strategic features (export and business performance)	
	External and internal growth	
	Ownership structure	Fleuren et al., 2014 Wu et al., 2002; Krasniqi and Kutllovi, 2008; Suwannaporn and Speece, 2010; Yang et al., 2015 Al-Mubarak et al., 2015 Astebro and Michaela, 2005 Fleuren et al., 2014 Fleuren et al., 2014 Fleuren et al., 2014 Yang et al., 2015 Escalfoni et al., 2011 Fleuren et al., 2014 De Brentanti and Kleinschmidt, 2004 Hemphälä and Magnusson, 2012 Fleuren et al., 2014
	Sector and country	
	Dimension	
	Formal structure	
	Cross-functional integration	
	Decentralized structure	
	Field of participation	
	Organizational architecture	
	Foreign ownership	
	-	
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Table 11 (continued)

Category	Indicator	Relevant references
Research and development activities and input	Internal communication	Lester, 1998; Suwanaporn and Speece, 2010
	Managerial decisions about which type of innovation to develop	Martinez-Ros, 1999
	Team and leadership	Cooper and Kleinschmidt, 1993; Cooper, 1999
	Good team structure together with appropriate leadership	Gana, 1992; Papadakis and Bourantas, 1998; Sorensen and Stuart, 2000; Blindenbach-Driessen and van den Ende, 2006; Slater et al., 2014; Raja and Wei, 2015
	Leaders/(senior) leadership/CEO characteristics/CEO change/senior management support	Lester, 1998; Blindenbach-Driessen and van den Ende, 2006
	Interdisciplinary teams	Blindenbach-Driessen and van den Ende, 2006
	External team communication	Bloch and Bugge, 2013
	Internal-management and internal-staff	Gerwin and Moffat, 1997; Hollenmann et al., 2009
	Team structure and characteristics	Griffin and Page, 1993; Weiss et al., 2011
	Team satisfaction and climate	Delaney et al., 1996
	Heterogeneity	Cavdar and Aydin, 2015
	National level	Freeman, 1979
	Gender distribution	Escalfoni et al., 2011
	Cyclic changes (birth, growth, and decline) and discontinuities in industry	Cooper, 1981; Cooper, 1999; Kivimäki et al., 2000; Henard and Szymanski, 2001; Parthasarthy and Hammond, 2002; Wan et al., 2005; Blindenbach-Driessen and van den Ende, 2006; Adams et al., 2006; Alcaide-Marzal and Tortajada-Esparza, 2007; Escalfoni et al., 2011; Kim, 2014; Nieves et al., 2014; Sosnowski, 2014; van Hoof et al., 2014; Song and Oh, 2015; Yang et al., 2015
	Handling of cultural differences of participants (level of education, experience, and others)	
	Input (people, physical and financial resources, tools)	
	Resource commitment	
	Equipment acquisition and maintenance	
	Presence of organizational resources	
	Business investment	
	Availability of equipment and tools/technologies	
	Maintenance/of resources	
	Availability of sufficient experts	
	Human capital stock	
	Human resources/human resources focused on innovation; Ratio of R&D D/employees	
	R&D personnel ratio	
	R&D intensity	
	Protected resources for non-core innovations	
	Project/company resource compatibility	
	Technological resource compatibility	
	R&D assets and strategies	
	Total cost	
	Project budget	
	Non-R&D innovation expenditures	
	R&D expenditure/investment/size of investment	
	Average expenditure per selected idea	
	Percentage of sales related to new projects	
	Development costs	
	R&D spending all/private	
	R&D investment and investors	
	Innovation expenditure per employee	
	Share of research budget from total company budget	
	Financial resource constraints	
	Technological investment	
	Extent to which the company management welcomes and supports research projects	
	R&D intensity	
	R&D budget	
	Innovation expenditure: investment, R&D, software, ICT expenditures,	

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Table 11 (continued)

Category	Indicator	Relevant references
Financial situation of the company	Project cost, Innovation financing, Share of technology transfer (of R&D), Innovation expenses, Share of R&D cost externally and internally, Design expenditure, Technology acquisition costs	Caird et al., 2013
	Resource efficiencies, e.g., reducing labor, materials, energy costs (innovation output)	Griffin and Page, 1993; Hippi and Grupp, 2005; Chiesa et al., 2009; Raymond and St-Pierre, 2010; Güngör and Gözli, 2012; Rocha et al., 2015
	(Increased) R&D activities/projects	Nilsson and Ritzén, 2014
	Active change of R&D working practices	Flor and Oltra, 2004
	Existence of formalized R&D in the company/formal allocation of resources and also the continuity of innovative effort within the business	Griffin and Page, 1993; Astebro and Michaela, 2005; Zdunczyk and Blenkinsopp, 2007
	Methods and tools	Beneito, 2003
	Financial autonomy	Martinez-Ros, 1999; Souitaris, 2002a; Yagüe et al., 2014; Cornaggia et al., 2015
	Financing/access to financing	Frey et al., 2013; Yang et al., 2015
	Access to capital market/capital management	Lecerf, 2012; Sosnowski, 2014
	Finance and support	Escalfoni et al., 2011; Bayarçelk et al., 2014
Financial performance based on innovation	Economic context during the process of innovation/economic situation	Cavdar and Aydin, 2015
	Stocks traded	Czarnitzki et al., 2011
	Taxes	Griffin and Page, 1993; Zahra, 1993; Tsai, 2001; Keizer et al., 2002;
	Improvement of performance/good/positive financial	Czarnitzki and Kraft 2004; Flor and Oltra, 2004; Astebro and Michaela, 2005;
	performance/performance related to innovative products/financial success based on innovation	Palmberg, 2006; Chiesa et al., 2009; Sawang, 2011; Idris and Trey, 2011; Caird et al., 2013; De Felice and Petrillo, 2013; Ivanov and Avastici, 2014;
		Dewangan and Godse, 2014; Kim 2014; Hittmar et al., 2015 (processed according to Trommsdorff and Steinhoff, 2007; Chromjaková and Rajnoha, 2009)
	Sales through innovation	
	Sales percent of new products and services in total sales	
	Sales under patent protection	
	Sales and sales growth	
	Share of sales with newly developed products	
	Share of turnover of new product	
	Share of a business' total sales deriving from innovative products	
	Share of new and improved products in total sales as output indicators	
	Share of new products in % from revenue = return on sales	
	Share of new products in % compared to the total revenue in comparison to previous years	
	Share of innovation expenses to total turnover	
	Revenue based on new products	
	Percent of new products that equals 80% sales	
	Profits under patent protection	
	Rate of turnover, perception of innovation returns	
	Revenue from new products, services, new customers	
	Innovation momentum: number of new products to sales	
	Volume of sales from innovative products	
	Payback period/amortization time	

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Table 11 (continued)

Category	Indicator	Relevant references
Market	Internal rate of return (%)	Cooper 1981 Hollenstein, 2003 Acs and Audretsch, 1987 Freeman, 1979; Zahra, 1993; Crépon et al., 1998; Astebro and Michaela, 2005 Cooper, 1981; Astebro and Michaela, 2005; Griffin and Page, 1993; Martínez-Ros, 1999; Huang et al., 2004; Astebro and Michaela, 2005; Belitz et al., 2011; De Felice and Pettilo, 2013; Bayarçelik et al., 2014; Liu et al., 2014 Cooper, 1981; Griffin and Page, 1993; Palmberg, 2006; Mendes Luz et al., 2015 Lukas and Ferrell, 2000; Ivanova and Avasilcăi, 2013 Kerssens-van Drongelen and Cooke, 1997 Astebro and Michaela, 2005 Baptista and Swann, 1998; Sáez-Martínez et al., 2014 Cooper, 1999; Kerssens-van Drongelen and Bilderbeck, 1999; Gonzalez-Benito et al., 2015 Griffin and Page, 1993; Fleuren et al., 2014; Astebro and Michaela, 2005; Chiesa et al., 2009; Enkel et al., 2005; Sawang, 2011; De Felice and Pettilo, 2013; Dewangan and Godse, 2014 De Felice and Pettilo, 2013 De Felice and Pettilo, 2013 Astebro and Michaela, 2005; Adams et al., 2006; Dewangan and Godse, 2014; Yang et al., 2015 Cooper, 1999 Griffin and Page, 1993 Hollenstein, 2003; Bayarçelik et al., 2014; Hittmar et al., 2015 Idris and Trey, 2011 Littrell, 1994 March-Chordà et al., 2002 Mendes Luz et al., 2015 Mendonça et al., 2004 Romero, 2014 Tohidli and Jabbari, 2012; Astebro and Michaela, 2005
	R&D costs/revenue in %, profit margin measures	
	New-to-market and new-to-business sales	
	Percentage increase in innovation revenues per employee, portfolio ROI realized	
	Current idea portfolio NPV/ROI/IRR (net present value/return on investment/internal rate of return)	
	-	
	-	
	-	
	Demand for growth in the industry	
	Duration of demand	
	Market share	
	Market share	
	Potential market size	
	Market position	
	Market size	
	New product introduction vs. competition	
	-	
	-	
	-	
	Customer complaints	
	Response time to customer requests	
	Support requests	
	Margin for customer	
	Customer profitability	
	Delivery	
	Delivery reliability and/or speed	
	Realization of innovation: Percentage of impact on customer satisfaction index	
	Customer retention rate	
	-	
	Loyalty rate	
	Rate at which new offerings are launched	
	Commercialization expenditure for the innovation portfolio	
	Rate of customer adoption of new offerings	
	Promotion cost	
	-	
	-	
	Sales share of new or highly improved services (%)	
	Cost reduction generated by process innovations (yes/no)	
	Number of new markets surveyed the (respective) last year	
	-	
	-	
	-	
	-	
	Constant introduction of innovations	
	Product distribution method that has been added to previous methods as a new one? (Tick in front of the method being added)	
	Number of sales agencies the company established inside or outside the province	
	Distribution channels	

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Table 11 (continued)

Category	Indicator	Relevant references
Use of new marketing methods	To what extent has the company used new marketing methods to sell its products over the past three years?	Tohidi and Jabbari, 2012
Customer need; customer requirements: cost, quality, security, interoperability, customer's financial strength	Need for cognition	Cooper, 1981; Wood and Swait, 2002; Astebro and Michaela, 2005; Yagüe et al., 2014
The intensity within the market	Need for change	Yang et al., 2015
Export (of high-technology)	High-technology exports	Martinez-Ros, 1999; Landry et al., 2002; Romijn and Albaladejo, 2002; Krasniqi and Kutllovi, 2008; Caird et al., 2013; Cavdar and Aydin, 2015
Exporting status	Export activities	
International orientation	International relations	
	Level of internationalization	
	International teams, multi-country market research and global or "local" products	Cooper and Kleinschmidt, 1993; Cooper, 1999; Landry et al., 2002; Romijn and Albaladejo, 2002; De Brentanti and Kleinschmidt, 2004; Güngör and Gözli, 2012; Frey et al., 2013
	Global orientation of product	
Propensity to work on foreign markets	-	Frey et al., 2013
Interaction of organization/between business' units	-	Lukas and Ferrell, 2000; Acs et al., 2002; Parthasarthy and Hammond, 2002; Romijn and Albaladejo, 2002
Internal and external collaboration	R&D alliances	Mansfield, 1998; Landry et al., 2002; Romijn and Albaladejo, 2002; Flor and Oltra, 2004; Koschatzky et al., 2001; Keizer et al., 2002; Blindenbach-Driessen and van den Ende, 2006; Belitz et al., 2011; Arvanitis et al., 2008; Oyelaran-Oyeiyinka and Adebawale, 2012; Caird et al., 2013; De Medeiros et al., 2014
University-industry collaboration, interfunctional collaboration, organizational support for inter-company external collaborations		
Collaboration with customers and suppliers and performing market research		
Participation in R&D projects with other organizations, universities, research institutes, competitors, consultants and service providers, suppliers/customers	Knowledge and technology transfer activities with research institution and/or institutions of higher education	
Connection to the local economic development	-	Pervan et al., 2015
Good network	Network: total size of strategic alliances	Sarvan et al., 2010
Collaboration with external partners	-	Astebro and Michaela, 2005
Connection with customers and user groups or user communities	-	Caird et al., 2013
Supplier relationship	-	Raja and Wei, 2015
Organizational support for collaboration with customer/user groups/user communities/spin-off innovative organizations	-	Caird et al., 2013
supplier linkages	-	Suwannaporn and Speece, 2010
Customer and supplier relationships	-	Kamasak, 2015
Cooperation with research centers	-	Frey et al., 2013
Collaboration and entrepreneurship	-	Sosnowski, 2014
Decision-makers	-	Beroggi et al., 2006
Autonomy for decision-making	-	Gana, 1992
Centralization of decision-making	-	Koberg et al., 1996
Innovative environment	Number of innovative businesses/new venture start-ups	Alcaide-Marzal and Tortajada-Esparza, 2007; Adams et al., 2013; Al-Mubarak et al., 2015
	Share of companies with innovations	Bloch and Bugge, 2013
	-	Brem and Voigt, 2009; Bloch and Bugge, 2013; Pervan et al., 2015
	-	Brem and Voigt, 2009
	-	Leydesdorff and Meyer, 2006; Brem and Voigt, 2009; Escalfoni et al., 2011; Yagüe et al., 2014
Stakeholders	-	Brem and Voigt, 2009
Political driving forces (e.g. government stability, taxation policy) and support by specific policies and programs	-	Brem and Voigt, 2009
Socio-cultural influences (e.g. income distribution, consumerism)	-	Gana, 1992; Caird et al., 2013
Environmental influences (e.g. protection laws), public control (government)/legal constraints or rules that restrict innovation process	-	Delaney et al., 1996
Economical influences (e.g. inflation)	-	
Technological influences (e.g. speed of technology transfer)	-	
External support	External financing (received by the company)	
Environmental monitoring (systematic monitoring of the developments that could affect it)	-	

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Table 11 (continued)

Category	Indicator	Relevant references
National level: national wealth, national economies	GDP per capita GNI per capita (PPP) Health expenditure (HE) Long-term unemployment R&D intensity (GERD/GDP) = GERD = Gross domestic expenditure on R&D National wealth (GDP per capita) Economy	Delaney et al., 1996; Katz, 2006; Leydesdorff and Meyer, 2006; Wang, 2012; Cavdar and Aydin, 2015; Pekovic et al., 2015
Pressure from outside: increased interest from retailers and consumers requiring accurate and relevant information on the performance and sustainability of products Reduced environmental and/or aspects of health and safety impact Regulations and standards Economic factor/economic structure		Van Hoof et al., 2014 Mendes Luz et al., 2015 Mendes Luz et al., 2015 Bayarçelik et al., 2014

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